

Monday, May 18th

Summary

- Training on the BSC and CL2

Next Steps

- LVL 0 Plasmid PCR and cleanup incomplete, finish tomorrow (2-3h)
- Training continued for other members tomorrow

Notes

BSC Training

- Rachel trained on BSC usage
- Cross contamination email from Marta [Cross Contamination Email](#)

EP-PCR

- Ran error-prone PCR with RNAt 1-3
- Results were as expected (show gel if there is one)
- Protocol: *Error Prone PCR (EP-PCR)* [Error Prone PCR \(EP-PCR\)](#)

Table 1. PCR parameters for EP-PCR

PCR Step	Starting Conditions
Denaturation	94°C for 1 minute
Annealing	60°C for 1 minute
Extension	72°C for 3 minutes

Table 2. Reagents used for EP-PCR

Reagent name	Amount
Water	51 µl

Tris Cl (pH = 8.3)	10 μ l
KCl	2.5 μ l
MgCl ₂	3.5 μ l
dCTP (pH = 7)	4 μ l
dTTP (pH = 7)	4 μ l
dATP (pH = 7)	4 μ l
dGTP (pH = 7)	4 μ l
5' Primer	2 μ l
3' Primer	2 μ l
Template DNA (400 bp in length)	10 μ l
MnCl ₂	2 μ l
Taq DNA polymerase	1 μ l

Footnotes

- PCR machine was making weird noises, informed Marta and she'll get back to us
- Running low on pipette tips
- etc.

Tuesday, May 19th

Summary

- Cathrine and Sabrina were trained on the BSC and CL2

Next Steps

- More members to be trained

Wednesday, May 20th

Summary

- Anthony, Anvi, Jacob and Emma were trained on the BSC and CL2
- Rachel and Melissa were trained on BSC, CL2, thermocycle, plate reader, and chemidoc

Next Steps

- Autoclave training when tips and media need to be autoclaved (Rachel and Melissa)
- Move in equipment and take inventory

Notes

BSC Training

- Book V6 or V7 when needed since V5 is being used by a lab working with LVL 2 pathogens

CL2 Training

- Lab coats are being decontaminated
 - Take off CL1 coat → go into CL2 → put on CL2 coat

General

- 70% ethanol 5min or 10% bleach 30min
- Book equipment on infinityX night before

Footnotes

- Camila and Andrea don't need training since they're on file from last year
- Jovita and Alex to get general training in June, Alex and Emma to get senior training in June.

Thursday, May 21st

Summary

- Cleaned up reagents and cultures from last year
- Took rough inventory of CL2

Next Steps

- Figure out how to extract glass beads from samples with glass beads (PEG, algae)
- Take detailed inventory of each box of DNA
- Determine if DNA kept at room temperature is still viable
- Update inventory tracker

Notes

Cleanup

- Separated old cultures and contaminated media from usable reagents and uncontaminated media
- Old cultures and contaminated media were soaked in a 10% bleach solution for about 30min, then cleaned and disposed of
- DNA kept at room temperature and cultures containing glass beads were kept aside
- Plastic culture flasks were set aside, might be disposed of later

Inventory

- Took note of contents of -80, -20, and 4 fridges
 - Pictures of boxes in -20 and -80 were taken since there were many tubes
 - Inventory from 4 has been noted
 - All notes are in a physical notebook

Footnotes

- Outstanding equipment to come in next week

Friday, May 22

Summary

- No experiments were done in the lab.
- Lab was continued to be organized

Next Steps

- Perform inventory in CL1 and CL2

Monday, May 25

Summary

- Did not go into lab

Next Steps

- Perform inventory in CL1 and CL2

Tuesday, May 26

Summary

- Collected items from Binjal
- Sorted plastics into the cabinet and the shelves, counted inventory for everything that is sorted
- Current layout
 - For the left cabinet:
 - Right side: pipette tips associated things
 - Top shelf: new tips
 - Bottom shelf: extra tips (not autoclaved), empty tip boxes, PCR trays, RNase free pipette tips
 - Centre to left:
 - Top: centrifuge tubes (not autoclaved), PCR tubes (not autoclaved), collection tubes, T25 flask, plates syringes, 50mL and 15mL tubes, cryotubes
 - Bottom: parafilm big roll, plastic containers, cryotube tray (with lid), pcr tube tray (with lid), 96 well plates, test tube caps
 - For the right cabinet:
 - Green foldable basket, disposable scalpel
 - For the right shelf: autoclaved P1000, P200, P10, centrifuge tubes, PCR tubes
 - For the left shelf: cut parafilm, test tube rack, styrofoam rack
 - At the very top of both shelves: empty cardboard boxes

Next Steps

- Clean out the items in the 2 cardboard boxes on the bench
 - The inoculation loops + plastics box may be better suited for c12
- One of the autoclaved PCR tubes box is broken because the lid isn't always working (located on the lab bench, orange box)
 - If possible, it may be better to take the PCR tubes and transfer to another box to re autoclave
- Binjal will come tomorrow with more reagents after 12 tomorrow

Wednesday, May 27

Summary

- Continued to move in lab equipment from last year
 - Plastics, pipettes, tubes, reagents
 - Took rough inventory of most items
- Broken equipment and unneeded reagents for HSI was set aside
 - Incompatible P1000 tips + broken amazon pipettes
 - Baking soda for buffer

Next Steps

- Continue to organize and take inventory

Thursday, May 28

Summary

- Cleaned up CL1
- Filled up boxes of pipettes to be ready for autoclaving
 - Will make a system later with a location for 'Things to be autoclaved' and a location for 'stocked autoclaved items' just to have a constant running system of sterile stock
- Labelled everything with the intended place. See Figure 1 for information regarding where equipment is kept
- Made 70% EtOH for lab bench
- Put a lot of more 'stock' supplies up at the top
- Potential rack idea
 - Binjal got ahold of these plastic racks, the idea is to see if we can sterilize them in the autoclave. Once autoclave training is complete, discuss with Marta if this is an option

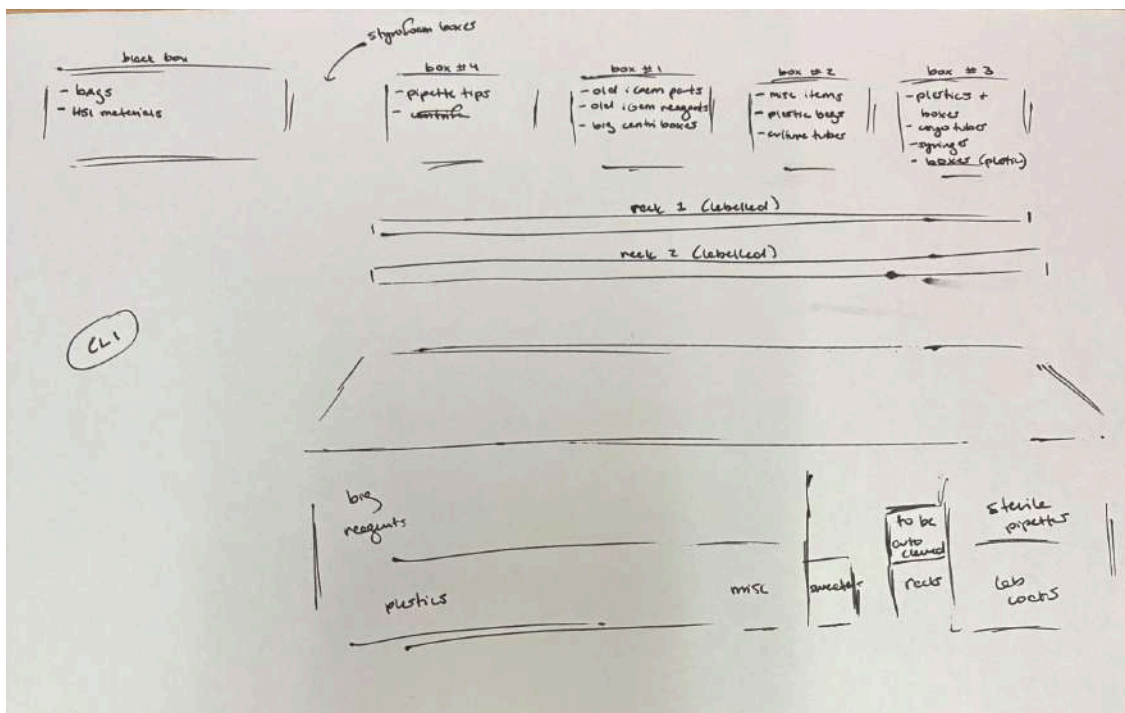


Figure 1. Visual depiction of how equipment has been organized in the lab.

Next Steps

- Made a couple piles on the lab bench as 'tasks' for the next people that come in
 - Primarily organizing CL2 and bringing extra materials in
 - Should be arranged similar to CL1 (with easy to grab things on the top, stock on bottom)
 - Sorting out the miniprep/purification kits (inventory of what is inside and what is missing from the kit. Fridge component should be in fridge in CL2 in 4 degree fridge on the side door (clear bin labelled synbio))
 - For clarification, just put a sticky note and write everything we have in the kit / search up the exact kit online and see what we're missing if possible!
 - Find pipette tips that match boxes / boxes that match tips.
 - First of all, using a p1000 from CL2, confirm that the tips in CL1 (in a beaker with Al foil sitting on lab bench) fits
 - If yes – go to lab store in ABB and see if there are corresponding tip boxes

Notes

- I know its a lot, if you need any help / have any questions feel free to contact Camila.

Friday, May 29

Summary

- CL2 was cleaned and organized
- P1000 tips were tested and do not work on our pipette
- Plasmid and PCR kits were reorganized and inventory was taken (reagents were mixed up between boxes). Extra purification kits were stored in Box #5 in CL2 above, small kits were taken to CL2

Next Steps

- Check with Marta regarding lab coats (seems that CL2 does not have any)
- Find correct pipette tips from the lab store (see point 2 under Summary section)
- Take inventory of everything, as there is now a general location of where things are going to be located

Monday, June 01

Summary

- Created reagents: MgCl_2 (stored in 4 fridge), CaCl_2 (stored in 4 fridge) and LB Broth (stored in CL1).
- Took brief inventory of CL2 bench space.
 - Took notes in notebook at the CL2.
- Got photo of P1000 tips at ABB (see figure 1 below)
 - Need to bring pipette in to check



Figure 1. Pipette P1000 tips.

Next Steps

- Need to autoclave reagents created
 - Note: MgCl_2 container is plastic (need to transfer).
- These reagents still need to be made:
 - LB Agar Base
 - LB + Ampicillin plates
 - 100 mM CaCl_2
 - 50% Glycerol
 - 1.0 M LiAc
 - 50% PEG 3350
- Need to purchase P1000 pipette tips & check pipette tips for empty boxes (located in CL1)

Notes

LB Broth Prep

- Made 500 ml
- Needs to be autoclaved
- Based on: *Solid LB Media Preparation* [Solid LB Media Preparation](#)

1M CaCl₂ Prep

- Made 100 mL of 1 M CaCl₂.
- Needs to be autoclaved
- Stored in 4°C fridge.
- Based on: *E. Coli Heat Shock Transformation* [E. Coli Heat Shock Transformation](#)

200mM MgCl₂ Prep

- Made 100 ml of 200mM MgCl₂
- Needs to be autoclaved
- Stored in 4°C fridge
- Based on: *Error Prone PCR (EP-PCR)* [Error Prone PCR \(EP-PCR\)](#)

Table 1. Summary of Reagents Created

Reagent	Amount
LB Broth	500 ml
1 M CaCl ₂	100 ml
200 mM MgCl ₂	100 ml

Footnotes

- LB Broth powder is running low.
- CL2 lab coats are still not available.

Tuesday, June 2

Summary

- Lab store:
 - Bought 1 bag of P1000 tips
 - Bought gloves – small, medium, large (1 box of each size)
 - Bought 1 L of Glycerol
- Made 100 mL 100 mM CaCl_2 (stored in CL2 4 fridge), 100 mL of 50% glycerol (on CL1 bench top)
- Picked up E. coli plate at 1:30 from Dr. Hoare, (stored in the CL2 4 fridge)
- Moved 2 lab coats to CL2
 - Marta is washing old lab coats (they are drying at the sink near the fridge in CL2)
- Updated inventory

Next Steps

- Make LB, plates + plates with amp in it
- Autoclave
 - Talk to Marta about training
- Need to buy PEG 3350 and LiAC

100mM CaCl_2 Prep

- Made 100 mL of 100mM CaCl_2 .
 - Took 10 ml of 1M CaCl_2
- Needs to be autoclaved/sterilized
- Stored in 4°C fridge.
- Based on: *E. Coli Heat Shock Transformation*  *E. Coli Heat Shock Transformation*

50% glycerol

- Made 100 mL of 50% glycerol.
- Needs to be autoclaved/sterilized
- Stored in CL1 countertop.

Table 1. Reagents created

Reagent	Amount
100mM CaCl ₂	100mL
50% glycerol	100mL

Footnotes

- Have lots of boxes of P1000 tips that fit P1000
- See image below. The box with the yellow post-it note has tips that do not fit which can be used for HSI. The box at the bottom of the picture has tips that fit



Figure 1. Image of P1000 tips that do not fit.

Wednesday, June 3

Summary

- Finished E. Coli Culture + Solid media prep.
- Autoclaved tips

Next Steps

- E. coli growth curve
- Prepare competent cells
- Plates stored in 4 degree fridge in CL2

Solid Media Prep.

- Made 2 Solid Media preps (500 ml each) according to *Solid LB Media Preparation protocol*. [Solid LB Media Preparation](#)
- One of them was made with only 12.3g LB powder (ran out), this batch was later used for the culture
- The other batch may have been contaminated
 - The tin foil was slightly shifted after the flask was autoclaved
- An extra batch was made because protocols said 15g of bacto-agar, whereas only 7.5g was needed per batch

E. Coli Culture

- Completed according to *E.coli Culture* protocol [E. coli Culture](#), no major deviations.
- The gel slightly broke during streaking process
- An extra plate was prepared (with no E. Coli) as a backup

Thursday, June 4

Summary

- Camila found yesterday's bacteria in the fridge
- Put in 37C incubator at 10am
 - Came back in at 4:30pm to check for growth, there was growth but bacteria streaks were too saturated (were streaks done using the same inoculation loop?). See figure below.

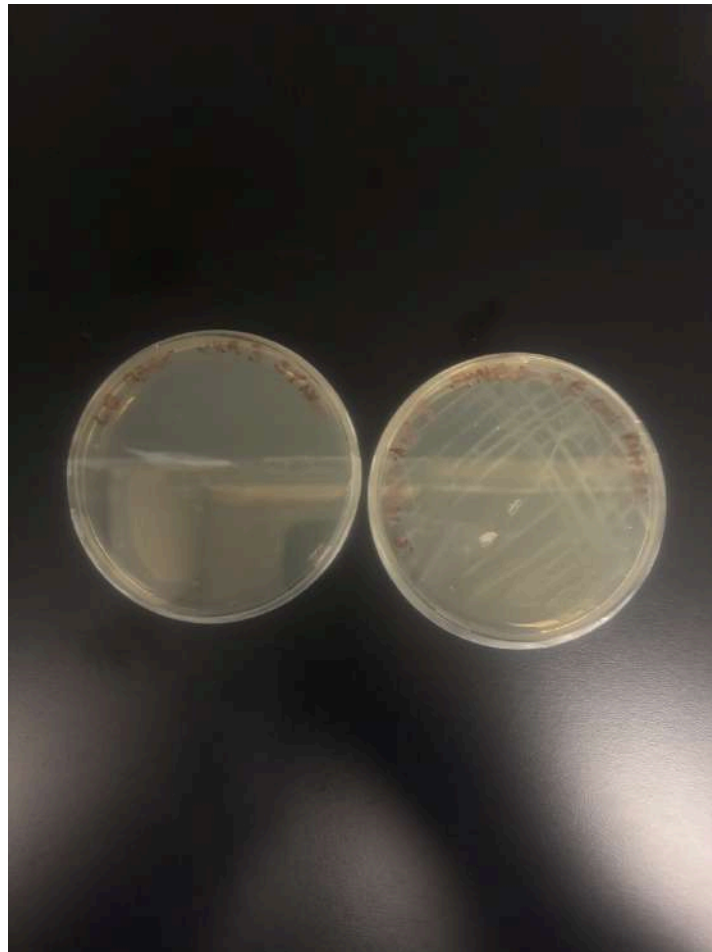


Figure 1. E.coli plates after incubation. Streaks appear saturated.

- Images were taken of the final setup of the CL2 bench to keep track of inventory location.
- Bacteria was unusable. Killed with 10% bleach solution

Next Steps

- Bacterial culture will continue next week. This includes making competent cells and growth curve
- Making antibiotic concentrations (to later make the plates)
 - 40mL stock solutions
 - Ampicillin and kanamycin (is kanamycin needed?)
 - See Figure 2 below for optimal concentrations.

Antibiotic	Recommended Stock Concentration	Recommended Working Concentration
Ampicillin	100 mg/mL	100 µg/mL
Blebbistatin	5 mg/mL	5 µg/mL
Carbenicillin*	100 mg/mL	100 µg/mL
Chloramphenicol	25 mg/mL (dissolve in EtOH)	25 µg/mL
Coumermycin	25 mg/mL (dissolve in DMSO)	25 µg/mL
Gentamycin	10 mg/mL	10 µg/mL
Kanamycin	50 mg/mL	50 µg/mL
Spectinomycin	50 mg/mL	50 µg/mL
Tetracycline	10 mg/mL	10 µg/mL

Notes:
Unless otherwise indicated, the antibiotic powder can be dissolved in dH₂O.
*Carbenicillin can be used in place of ampicillin. Carbenicillin is more stable, so it is potentially more effective at selecting only bacteria containing the plasmids of interest (for example, fewer satellite colonies will grow). However, it is also more expensive.

Figure 2. Optimal concentrations of antibiotics taken from the iGem website.

- Calculate the amount of powder needed to make the concentration (using either EtOH or milliQ water)
 - Send the calculations to Camila or Rachel to confirm before proceeding
- Make the solution in the BSC
- Filter the solution through a syringe
 - Will likely need to buy the syringes and filters from lab store
 - 0.22 µm filters (check if there are some in the CL1 lab bench cabinet)
 - 15mL–20mL syringe (always want to use a bigger size than solution being worked with)

Friday, June 5

Summary

- Went to lab store
 - PTFE Syringe Filter (Diameter (mm) = 25 . Pore Size (um) = 0.22
 - Luer Lock Syringe - without needle (20 mL volume)
- Concentration calculated of ampicillin stock solution - see figure below

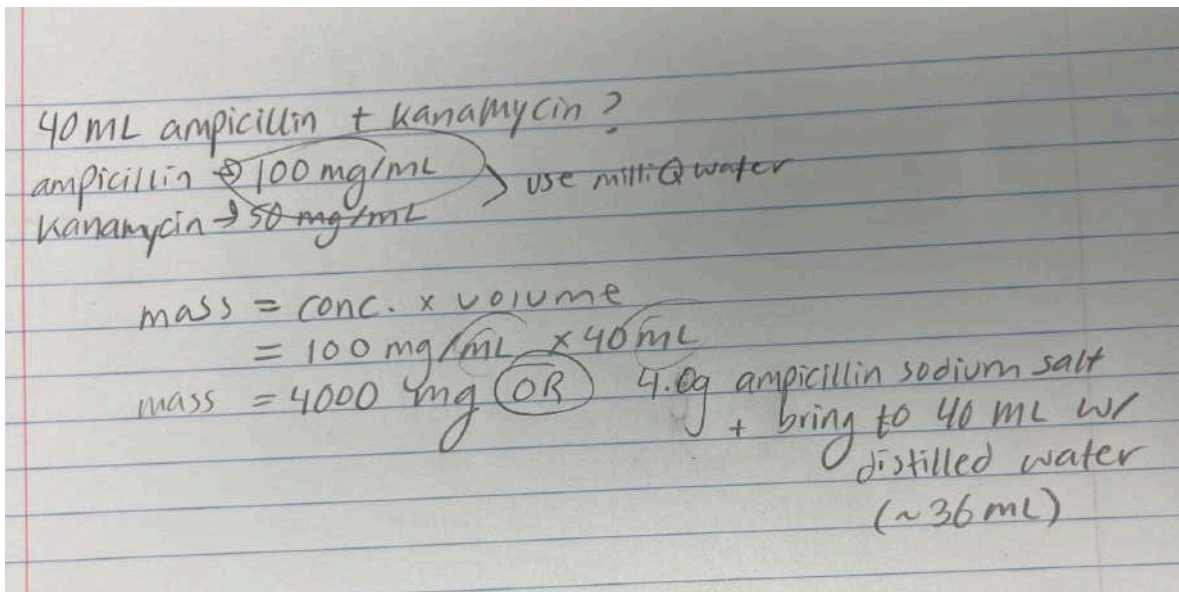


Figure 1. Calculated concentration of the ampicillin stock solution

- Ampicillin 100 mg/mL recommended stock concentration created (40mL)
 - Gathered 4g of Ampicillin → Filled to 40mL with milliQ H2O
 - Used syringe filter to sterilize and vortexed until clear
 - Pre-wet the filter with ethanol and then milliQ water
 - Stored in CL2 4 fridge

Next Steps

- Could use another black sharpie in CL2
- Unclear of next steps at this point → Ask Camila / Rachel

Notes

- Erlenmeyer flask left on drying rack far left side
- Opened autoclaved p1000 box in CL2 & used 2 tips (left tape on top)

-
- Ampicillin solution created on bench and not within the BSC however, everything sprayed thoroughly with ethanol
 - Lab inventory updated with new items from lab store

Monday, June 8

Summary

- Poured one agar plate (it was a bit chunky).
- Streaked E. Coli and placed in a 37-degree incubator (In the corner on the bottom one)

Next Steps

- Need to check the incubator for E. coli solid prep
- Start E. coli liquid culture

Notes

E. Coli Streaking

- Poured one agar plate
- Streaked E. coli using 2 inoculation loops
- Sealed with parafilm, labelled, and incubated at 37 degrees, inverted
- Based on this protocol: *Solid LB Media Preparation* [Solid LB Media Preparation](#)

Footnotes

- Marta said that the temperatures of the water bath can be increased (does not have to be set at only 37 and 60 degrees C).
- 30 minutes after autoclaving, try pouring plates the same day because reheating LB broth with agar is time-consuming.
 - The agar plate poured was not fully liquid (could not wait longer to complete)
- Agar melts at 80 degrees.

Tuesday June 9

Summary

- The plates made yesterday looked very streaky
 - Found 2 colonies
- Made liquid inoculations of the DH5a E.coli
 - 50ml falcon tube: 25mL LB + e.coli (colony 1)
 - 15mL falcon tube: 5ml LB + e.coli (colony 1, which is the same colony from the 50mL falcon tube)
 - 15mL falcon tube: 5ml LB + e.coli (colony 2)
- 50 falcon can be used for growth curve
- 15ml falcon tube can be used competent cells (choose whichever replicate is better)
- All the tubes with the inoculations are in the bottom incubator, far left in the corner
- Tubes are in 37 degrees, 250rpm shaking, loosely capped and taped
- Followed this protocol: *E.coli Culture* [E. coli Culture](#)



Figure 1. E.coli plate from yesterday, appears streaky.

Next Steps

- Autoclave the 50% glycerol (need it tomorrow for competent cells)
- Still need to sterilize the 100mM CaCl₂
- Take growth curve and make competent cells

Footnotes

- Pipette gun is charging on CL1 bench
- Don't use autoclave tape for labelling
- When labelling, add initials of who made it
- Ask Marta about lab coats drying last week

Wednesday, June 10

Summary

- Autoclaved a few things (specifics below)
- Made more liquid LB
- Gathered data for the OD curve
- Reassured inoculations (diluted 1:100 LB)
 - 4.95LB 50mL inoculations
- Brought cuvettes (2 unopened with 100 each + opened box with 89 – opened one was brought into CL2)
- Re-passaged liquid culture for CC

Next Steps

- Component cells (did not have time)
- Consider re-doing the growth curve for more data
- Find a way to autoclave 70mL culture tubes
- Buy 100ml flasks if needed

Notes

Autoclaving

- 100 mM CaCl₂
- 1 mol CaCl₂ 100mL
- 50% glycerol 100mL
- Liquid LB 500mL
 - Made this today following same protocol
 - Protocol: *Solid LB Media Preparation* = Solid LB Media Preparation
 - Needed more stuff to autoclave + needed more LB

OD Curve

- Did not have 100mL flasks and *Optical Density Monitoring*
 - = Optical Density Monitoring protocol called for 100mL 1:100 dilution of overnight liquid culture

- Did 1:50 dilution of 45mL of LB instead to allow for shaking and aeration in falcon tube (and leave at a reasonable time)
- 90uL ovn culture + 44.1mL LB, + 25mL LB placed in incubator for blanks

Table 1: OD Curve Incubation Log

Timepoint	Incubated	Removed
1	12:55	1:25
2	1:45	2:15
3	2:23	2:50
4	2:55	3:25
5	3:35	3:55
6	4:05	4:25
7	4:32	4:53
8	4:58	5:17
9	5:21	5:42 (placed in fridge).

Table 2: Optical Density Curve Data

Time (measured)	1 (Blank)	2	3	4	Average
1:35	0	0.004	0.045	0.006	0.018
2:20	0	0.093	0.096	-	0.0945
2:55	0	0.172	0.166	-	0.16
3:33	0	0.284	0.286	-	0.285
4:05	0	0.385	0.386	-	0.3855
4:30	0	0.487	0.489	-	0.488
5:00	0	0.606	0.604	-	0.605
5:22	0	0.671	0.669	-	0.670
5:46	0	0.765	0.771	-	0.768

-
- Trial 1 used 1mL each (not enough volume) – needed to do (4) to troubleshoot
 - Values did not make sense, could be due to steam from hot LB and also low liquid volume
 - Trial 2 onwards used 2mL, readings more clear
 - 1:100 re-dilution for competent cells (15mL falcons) and OD cells
 - 50uL culture + 4.95mL LB for both replicates
 - Came in early Thursday to do 1:100 dilution for CC (1:15pm)

Thursday, June 11

Summary

- Made a 85% calcium chloride 15% glycerol solution using 1 M CaCl_2 and 50% glycerol solution
 - Calculation below

Handwritten calculations for a 85% calcium chloride, 15% glycerol solution.

Method 1:

$$C_1V_1 = C_2V_2$$

$$\frac{(200\text{mM})(x)}{1000} = \frac{(85\text{mM})(100\text{mL})}{1000}$$

$$x = 42.5$$

Method 2:

$$C_1V_1 = C_2V_2$$

$$\frac{(80\text{mM})(x)}{100} = \frac{(15)(100)}{100}$$

$$x = 18.75$$

Sum we don't have: 500mM

Method 3:

$$(1000\text{mM})(x) = (85\text{mM})(100\text{mL})$$

$$x = 8.5\text{mL}$$

Method 4:

$$(1000)(x) = (85)(50)$$

$$x = 4.25\text{mL}$$

Final volumes:

45.2mL 200mM CaCl_2	22.6mL
30mL 50% glycerol	15mL
24.8mL water	12.4mL
= 100mL	

divide everything by 2

we don't have anything anymore

in that can store 100mL

Final volumes (divided by 2):

4.25mL 1M CaCl_2	
15mL 50% glycerol	
80.75mL water	

Figure 1. Calculations for 85% calcium chloride, 15% glycerol.

- Made 40 DH5 alpha competent cell solutions in cryotubes using [this](#) protocol
 - Initial OD for both 50 mL diluted bacteria tubed (Rachel came in at noon to dilute 1:49), was 0.543 for replicate 1 and 0.497 for replicate 2
 - Stored in -80 deg C
 - Stored on the TOP 2-2 rack in the -80 deg freezer
 - Aliquotted amounts have 200 uL in each
 - They are separated from colony 1 and 2 (both from same plate, to differentiate (eg if we have any issues in future) colony 1 has "June 2026" written down, whereas colony 2 has "June 10/26" written)
 - Both colonies should be fine, just made differentiation in case we were having issues with one or the other
 - Diluted 3 samples for growth curve on Friday
 - Centrifuged and resuspended pellet in 1-5 mL of LB broth

- For 15 mL falcon tubes, took 50 uL ovn culture and put in 5 mL of LB (2x)
 - For 50 ml falcon tube, took 250 uL ovn culture and put in 25 mL of LB (1x)
 - They were both left in shaking incubator (bottom) at 27 deg C at around 250rpm
- Accidentally made an extra falcon tube with the 1 M CaCl₂ solution in it. Just use as an aliquotted amt

Next Steps

- Growth curve again on Friday!
- Potentially aliquotting media? (I want to be able to heat up before dilutions, reheating the big glass bottle is tricky – also dont love bringing in big stock from fridge into bsc)
- Pls remove the waste I have sitting in CL2 bench (bleached bacteria, dump out in sink and run water for 20m, dump solid waste in biohazardous waste)
- OUR P1000 TIPS DONT HAVE FILTERS???
 - A task for next week ;-; we need to find some from lab store (or have a look to see if the ones in box above have filters, and if they fit in boxes we currently have)
 - Filtered tips cant be autoclaved nvm, we need to buy them individually. Get samples from lab store and try with pipette?
- Fill MilliQ water for CL2 (have small aliquotted amt)
- Pls bring falcon tubes from CL1 into CL2 (we are running out)
 - They are located in bottom cabinet in CL1
- Binjal is collecting things
 - TAP should be in fridge (4 degree) in CL2, in white translucent bin on the ride of the fridge (where you put amp))
 - Not cure what she means by cc450? We have algae in the white styrofoam box in bottom cabinet of CL2, is mixed in with samples with beads. There should be centrifuge tube with JUST algae, not all have beads

Notes

- Vortexed at a low speed to break up pellet (would not break up AT ALL) with pipette mixing
 - This is not recommended,, look into potentially using less rpm in centrifuge? Pellet was far too dense. Also next time instead of doing this in 50 mL falcon tubes, should do it in 15 mL (pellets were too big to break up)

-
- As we are questioning viability, we should do a transformation TEST next week (puc19 plasmid in freezer as control?)

Friday, June 12

Summary

- Today we did growth curve again with the goal of getting more points on the curve (e.g taking it past log phase if possible)
- Protocols referenced can be found here: *E. coli culture* [E. coli Culture](#) and here *Optical Density Monitoring* [Optical Density Monitoring](#) .
- Did the same 1:50 ratio as they did on Wednesday (44.1mL LB, 0.9mL Culture)
- We did not have 100mL flasks (used 50mL falcon tubes)

Next Steps

- Stocking solutions for transformation.

Notes

OD Curve

Table 1: OD Curve Incubation Log

Timepoint	Incubated	Removed
1	12:05	1:00
2	1:20	1:58
3	2:06	2:32
4	2:36	3:00
5	3:05	3:29
6	3:35	4:00
7	4:04	-

Table 2: Optical Density Curve Data

Time (measured)	1 (Blank)	2	3	Average
1:27	0	0.105	0.103	0.104
2:08	0	0.247	0.241	0.244
2:41	0	0.337	0.335	0.336
3:06	0	0.462	0.463	0.463
3:35	0	0.577	0.584	0.581
4:05	0	0.656	0.658	0.65

Instructions (from Rachel):

Try to line up the arrow on the cuvettes with the rectangle in the well (one side will be diff) and wipe gently with a kimwipe

You've done this in Chem so it should be similar

Then use sample to select well "1" which is the blue well, and hit set ref

Then it'll read 2, record after the number stops changing

Then hit run and it'll read 3, same deal

Figure 1. Image of instructions for OD Curve.

Footnotes

- Small flasks labelled "DO NOT USE" and placed in lower cabinet in CL1
 - Binjal stated it was used last year so resultantly not useable!
 - She said they are not sterile
- Put the remaining E.Coli and LB 1:50 solution back in the incubator alongside the 1:100 solution (may need to be discarded... not sure what we are planning on doing with those yet)

Monday, June 15

Summary

- Made 500 ml of LB Agar broth
- Made 250 ml of LB broth
- Poured LB broth from 4C fridge into falcon tubes
- Autoclaved things (discussed below)
- Bought 1 pack Falcon tubes 50mL from the lab store.

Next Steps

- Start puc19 plasmid transformation into competent cells and plates.

Notes

LB Agar broth

- Made 2 beaker of LB Agar broth (250 ml each)
- Autoclaved them
- Followed this protocol: *Solid LB Media Preparation*  Solid LB Media Preparation
- Stored in 4C

LB broth

- Made 1 beaker of LB broth (250 ml)
- Autoclaved it
- Followed this protocol: *Solid LB Media Preparation*  Solid LB Media Preparation
- Stored in 4C

Transferred LB broth from fridge

- Melted LB broth in 4C fridge using communal water baths
 - Marta let us set to 85C
- Poured into falcon tubes in BSC
- Stored in 4C fridge



Figure 1. LB broth in falcon tubes stored in 4C fridge.

Autoclaved

- 250 ml of LB Agar brother x 2
 - Stored in 4C
- 250 ml of LB broth x 1
 - Stored in 4C
- 500 ml beakers x 3
 - Stored at CL2 bench



Figure 2. Autoclaved materials in 4C fridge in CL2.



Figure 3. Empty autoclaved bottles on CL2 shelf.

Falcon tubes

- Purchased one pack from lab store
- Stored in CL1



Figure 4. Falcon tubes stored in CL1.

Tuesday, June 16

Camila notes

- Make plates beforehand (1st thing you do when you get to lab! Put LB agar on hot plate)
 - 2x LB agar + amp plates (to see if transformation worked)
 - LB agar amp = transformed bac
 - LB agar amp = untransformed comp cells
 - 2x LB agar plates (to see if something happened to bacteria)
 - LB agar = transformed bac
 - LB agar = untransformed comp cells
- There are 200 uL in each comp cell vial, use 200 uL for transformation and 200uL to evaluate untransformed bac viability (two tubes). Just put the untransformed 200 uL in LB broth and grow in incubator to later put on plates
- Andrea do you remember if we did waterbath last year? I thought we had a mini oven thing in CL2
- Make sure to leave either <100 uL so we can do 2 plates per transformation (+ the 2 untransformed plates)

Summary

- Made 4 plates
 - 2 LB agar plates
 - 2 LB + 100ug/mL Ampicillin plates
 - Took 75mL of LB agar and added 75uL of 100mg/mL AMP

Calculations for making LB+AMP (100 µg/ml) agar plates

Amp stock = 100 mg/ml
↓
100 µg/ml

$$100 \mu\text{g/ml} \times 1000 \times V_{\text{amp}} = 100 \text{ ml (100 } \mu\text{g/ml) LB agar}$$

$$V_{\text{amp}} = 0.1 \text{ ml} = 100 \mu\text{l}$$

$$100 \mu\text{g/ml} \times 1000 \times V_{\text{amp}} = 75 \text{ ml (100 } \mu\text{g/ml)}$$

$$V = 0.075 \text{ ml} = 75 \mu\text{l} \leftarrow \text{used this}$$

Figure 1: Calculations for making LB and AMP

- For the transformation
 - Added 5ul of plasmid
 - Plasmid concentration is 50pg/ul → 50pg/ul x 5ul = 250pg (in the range)
 - Followed *E.coli* Heat Shock Transformation
 - ▢ E. Coli Heat Shock Transformation
 - 45 seconds of heat shock
 - Control was made by taking 200ul of competent cells, adding 950ul LB, incubating
- Took competent, did transformation with pUC19, plated them
 1. AMP plate + plasmid
 2. LB plate + plasmid
 3. AMP plate + no plasmid (control)
 4. LB plate + no plasmid (control)
- Marta gave lab coats, it is in CL1 bench currently

Next Steps

- Count colonies, make sure controls work

Notes

- Marta says: When turning off BSC, do not turn off blower



- ← this needs to be always on/green light
(Figure 2. Icon of blower on BSC).
- If using the CL2 microcentrifuge and the door is not opening after using it, there is a small unwinded paperclip on the side of the machine
 - Push the paperclip down in the small hole that is on the centre of the machine, above the screen, below the door
 - Push a little hard, should hear a click sound for it to open

Wednesday, June 17th

Summary

- Inspected agar plates for overnight growth
 - Placed in 4C fridge afterwards to slow growth
- Counted colonies on LB+AMP E.coli DH5alpha + pUC19 plate
- Prepared Liquid LB + AMP media
- Prepared two 5mL overnight cultures using two colonies from the plate
 - Placed in 37C shaking incubator

Next Steps

- Miniprep liquid overnight cultures
- Measure using DNA concentration of liquid overnight cultures

Notes

Colony counting

- Expected results were observed on control plates
 - Lawn on transformant plate and non-transformed plate without antibiotic
 - No growth on non-transformed plate with antibiotic
 - Colonies on transformant plate with antibiotic
- Plates were placed in the 4C fridge after counting to slow growth
- Colonies on transformant plate with antibiotic were difficult to count, likely because the transformants were not spread evenly across the plate
 - 32 colonies were counted outside of the 'blob' in the centre
 - About 69 colonies were counted inside the 'blob' in the centre

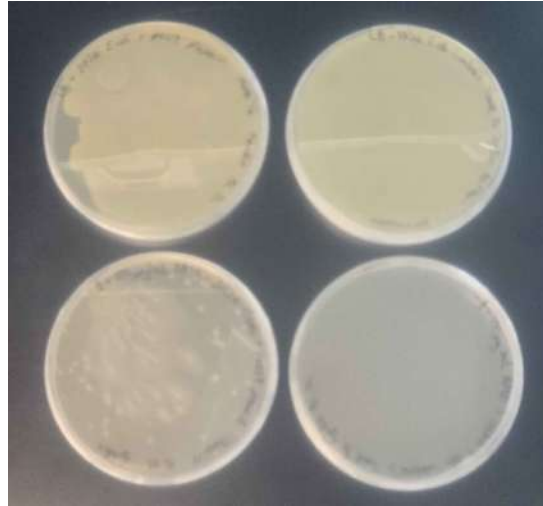


Figure 1. Photo of overnight growth on agar plates. From top to bottom, left to right, LB + E.coli DH5alpha + pUC19, LB + E.coli DH5alpha, LB + AMP + E.coli DH5alpha + pUC19, LB + AMP + E.coli DH5alpha. A bacterial lawn is observed in the top two plates and no growth is observed in the bottom right plate. Several colonies are observed in the bottom left plate.



Figure 2. Photo of overnight growth on LB + AMP + E.coli DH5alpha + pUC19 plate. A large, spreading blob in the centre suggests that the transformation solution was pipetted on the plate and not spread afterwards.

Liquid overnight culture

- Prepared 100ug/mL dilution of liquid LB + ampicillin in BSC (did not want to contaminate LB stock)

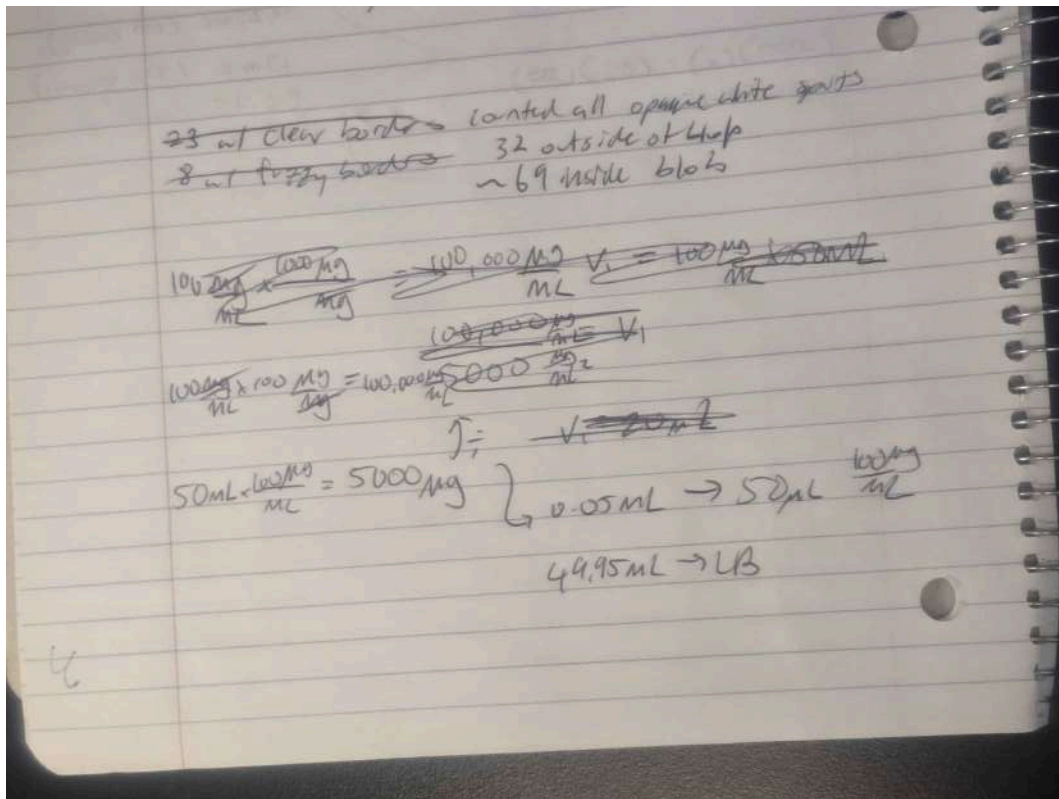


Figure 3. Image of calculations for dilution of LB liquid.

- Inoculated 5mL LB + AMP with one colony, 2x
- Placed inoculations in the 37C incubator with shaking
- Extra reagents were placed in the 4C fridge

Table 1. Reagents used for liquid culture preparation (50mL falcon tube)

Reagent	Amount
100mg/mL Ampicillin	50uL
Liquid LB	49.95mL

Table 2. Reagents used for liquid culture inoculation(15mL falcon tube each)

Reagent	Amount
100ug/mL LB + Ampicillin	5 mL (x2)

Colony from LB + AMP + E.coli DH5alpha + pUC19 plate	1 (x2)
--	--------

Footnotes

- Picked up P200 and P1000 RNase free tips from labstore (2 boxes each)



Figure 4. New pipette tips P200 and P1000 from the lab store.

Wednesday, June 18

Summary

- Minipreped both samples left (taken from two colonies)
 1. Both were from LB + 100 ug/uL amp liquid cultures
 2. Noticed colony 1 had less noticeable precipitation and was overall less pronounced during the miniprep process compared to colony 2
- Nanoquant was used to analyze plasmid concentration of the two minipreped samples
 1. Samples were then disposed of
- Dilutions in the BSC were bleached and left on CL2 counter to later dispose of

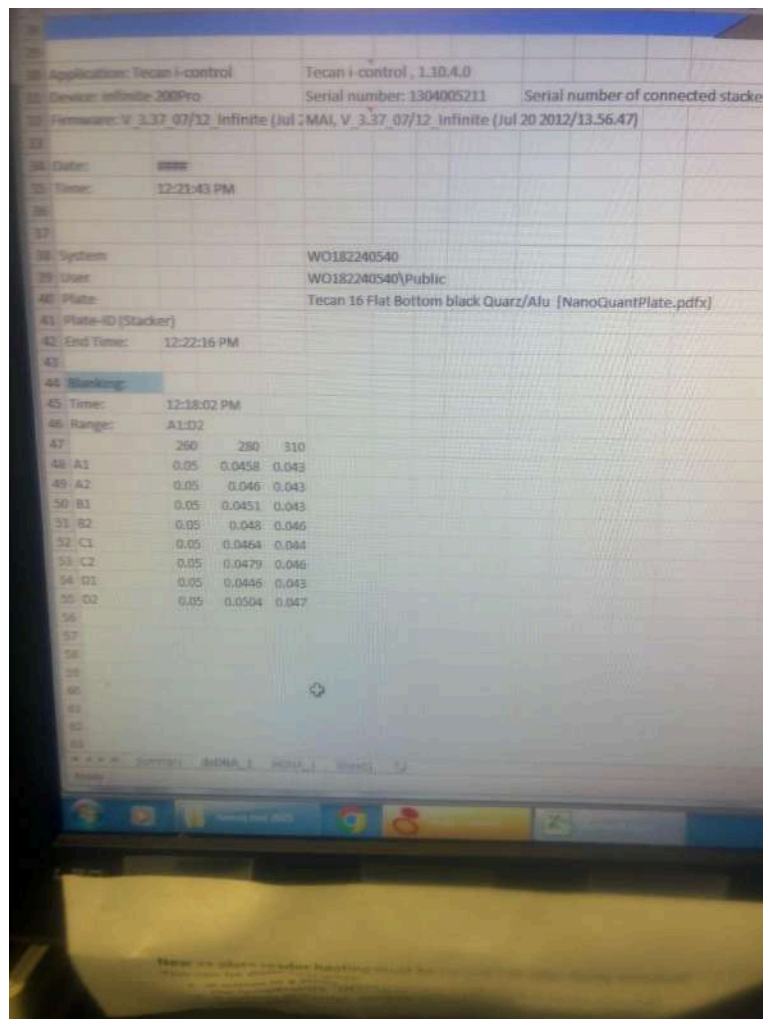


Figure 1. Calculations.

	1		2	
	Abs	Value	Abs	Value
A	260 0.1189 OD	118.9 ng/ul	260 0.1176 OD	117.6 ng/ul
	280 0.0632 OD	1.88 ratio	280 0.0618 OD	1.9 ratio
B	260 0.1202 OD	120.2 ng/ul	260 0.1171 OD	117.1 ng/ul
	280 0.0634 OD	1.9 ratio	280 0.0624 OD	1.88 ratio
C	260 0.0689 OD	68.9 ng/ul	260 0.068 OD	68 ng/ul
	280 0.0356 OD	1.94 ratio	280 0.0353 OD	1.93 ratio
D	260 0.0692 OD	69.2 ng/ul	260 0.0682 OD	68.2 ng/ul
	280 0.0352 OD	1.97 ratio	280 0.035 OD	1.95 ratio
E	260 OD	ng/ul ratio	260 OD	ng/ul ratio
	280 OD	ratio	280 OD	ratio
F	260 OD	ng/ul ratio	260 OD	ng/ul ratio
	280 OD	ratio	280 OD	ratio
G	260 OD	ng/ul ratio	260 OD	ng/ul ratio
	280 OD	ratio	280 OD	ratio
H	260 OD	ng/ul ratio	260 OD	ng/ul ratio
	280 OD	ratio	280 OD	ratio

Application: Tecan i-control Tecan i-control , 1.10.4.0
 Device: Infinite 200Pro Serial number: 1304005211 Serial number of connected stacker:
 Firmware: V_3_37_07/12_Infinite (Jul 12 MAI, V_3_37_07/12_Infinite (Jul 20 2012/13:56:47))
 Date: mmm
 Time: 12:21:43 PM

Figure 2. Calculations.

Next Steps

- Calculate transformation efficiency

Transformation Efficiency

amt of DNA = 250 pg 1 picogram = 0.001 ng
250 pg = 0.25 ng

TE = $\frac{\text{number of transformations}}{\text{amt of DNA transformed}}$ → $101 \cdot \frac{200 \mu\text{L}}{100 \mu\text{L}} = 202$

0.00025 ng amt of colonies counted amt of cells plated

TE = $\frac{202}{0.00025} = 8.08 \cdot 10^5 \frac{\text{CFU}}{\mu\text{g}}$

could be higher, counted dense region on the plate

Copy # copies per cell = $\frac{\# \text{ of plasmid mc recovered}}{\# \text{ of cells}}$

Plasmid DNA recovered (concentration) · (elution volume)

avg per colony 1 = $118.45 \text{ ng}/\mu\text{L} \cdot (30 \mu\text{L}) = 3553.5 \text{ ng}$
2 = $68.6 \text{ ng}/\mu\text{L} \cdot (30 \mu\text{L}) = 2058 \text{ ng}$

Figure 3. Transformation efficiency calculations.

Notes

- Is willing to give Andrea free Nanoquant training

Monday, June 22

Summary

- Made 7 LB + Amp plates & 8 LB plain plates
- Autoclaved 6 glass bottles
- Confirmed materials needed for gel electrophoresis

Next Steps

- Transform plasmids
- Look for DNA electrophoresis stain.

Notes

Plate preparations

- Made 7 LB + Amp plates & 8 LB plain plates
- Based on this protocol: *Solid LB Media Preparation* [Solid LB Media Preparation](#)
- Stored inverted in 4°C fridge



Figure 1. Plates in 4°C fridge.

Autoclave

- Autoclaved 6 glass bottles
- Left on C12 shelf



Figure 2. Autoclaved bottles on C12 shelf.

Footnotes

- Check for DNA stain needed for gel electrophoresis

Tuesday, June 23

Summary

- Andrea got nanoquant training
- Took take parts from 2025 plate 2 iGEM kit
 - M6 and A12
 - Resuspended at it is in CL2 -20, neon yellow box
 - For resuspension: added 10ul of water, resuspended
- Transformed parts into competent cells
 - For samples with the part, 1ul of the resuspended part was added
- Did heat shock for 45sec
- Plated on LB plates for no plasmid, and parts
- Plate on LB + AMP plates for no plasmid and parts
- total 6 plates
 1. AMP plate + M6
 2. AMP plate + A12
 3. AMP plate + no plasmid
 4. LB plate + M6
 5. LB plate + A12
 6. LB plate + no plasmid
- Gel electrophoresis ladder + purple dye is in -20 freezer in white box

Next Steps

- Check for colonies and inoculate
- If it looks suspicious, you may have to redo the transformation (add 1ul of plasmid to 200ul of competent cells) and make new plates
 - If re doing transformation, do 45 sec heat shock NOT 1.5min

Footnotes

- Make sure when pouring plates, the solution has not solidified yet
 - If it is just LB agar, just reheat it again and then pour
 - Clumpy plates = not good
- For labelling, if you are adding an antibiotic, please indicate what the concentration of the plate is

Wednesday, June 24

Summary

- Observed results of yesterday's transformation
- No growth observed on all plates with ampicillin
- Poured 2 lb plates from remaining agar
- Poured 6 lb + amp plates from 250mL LB agar stocks
- Re-attempted heat shock transformation

Next Steps

- Observe results + prepare overnight liquid culture for miniprep if successful
- Get P2 RNase free tips from LA store

Notes

Observing results

- Took plates out of 37 incubator and placed in fridge after observation
- All LB plates (DH5alpha, A12, and M6) had a bacterial lawn
- All LB + AMP plates (DH5alpha, A12, and M6) showed no growth
- Suspect it is due to lumpy agar plates (maybe ampicillin didn't distribute evenly) or improper transformation, so will reattempt both

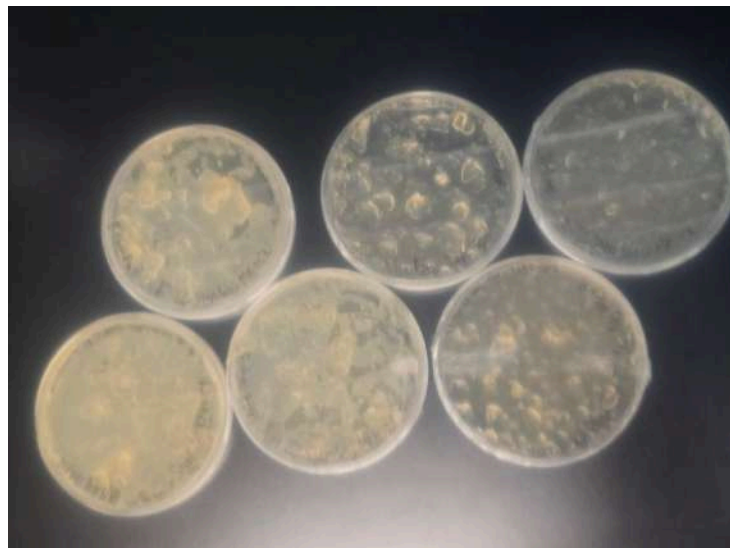


Figure 1. Agar plates, half have a lawn of bacteria and half have no growth

Pouring plates

- Heated agar using hot plate + beaker in water
 - Fairly fast (~45min) but stir rod would be preferable
- Used 0.025g ampicillin (halved amount for 500mL protocol since 250mL was prepared)
- Poured 2 LB plates from small flask and 6 LB + AMP plates from 250mL solution
- Some clumps due to lack of stirring but overall less clumpy

Table 1. Reagents used for LB + AMP plates and LB plates

Reagent	Amount
LB agar stock	250mL + ~80mL
Ampicillin	0.025g
Plates	6 + 2

Heat Shock

- Followed heat shock protocol (and edited to be 45s)
- Could not find P2 tips, noticed that P20 was set to 1uL (outside of range)
 - Will buy P2 tips tomorrow from labstore
 - Took about 10min to conclude we didn't have P2 tips, so 3 tubes of CC had to be discarded since they sat at RT for too long
 - Used 2uL of plasmid with P20
- Had to transfer to micro centrifuge tubes (1000ul) to spin, did 13.0 for 10s accidentally
- Plated cells on new plates + one LB plate from Monday
- Incubating at 37 degrees

Footnotes

- Will get P2 tips tomorrow (2 boxes)
- Will move the RNase free tips to c12 probably

Thursday, June 24

Summary

- Miniprepmed colonies.

Monday, June 29

Summary

- Lab cancelled for further lab planning.

Tuesday, June 30

Summary

- Camila got 1 p20 pack and 1p10 pack filter tips
- Took take parts O8 from 2025 plate 2 iGEM kit to see if kit is good
 - Resuspended at it is in CL2 -20, neon yellow box
 - For resuspension: added 10ul of NF water, resuspended
- Transformed parts into competent cells
 - For samples with the part, 1ul of the resuspended part was added
- Did heat shock for 45sec
- Plated on LB plates for no plasmid, and parts
- Plate on LB + AMP plates for no plasmid and parts
- total 4 plates
 1. AMP plate + O8
 2. AMP plate + no plasmid
 3. LB plate + O8
 4. LB plate + no plasmid
- Made extra 4 lb plates
- Did everything with filter tips

Next Steps

- See if there are colonies
- Buy plates and tape (look at notes)

Notes

- Need more plates, there's no more in Box 3
- Is the pink tape done? White tape was taken to cl2
 - Need more tape!
- LB made on June 1 in CL2 fridge looks cloudy, don't use it

Thursday, July 2

Summary

- Analyzed plates, saw negative results, will further troubleshoot
 - Reach out to Binjal? She might know more about distribution kits
- Cleaned up space
 - Put duplicate pipettes in storage
 - Organized tips, brought some of the RNase free ones to CL2 (they were hidden)
 - Disposed of plates that weren't needed in CL2 / liquid bacteria that was no longer needed
 - Started disposing parts from last year (in -20) that we don't need anymore
 - Got rid of dirty LB broth
- Took inventory of everything that we currently have (moreso reagents)
- Made more LB
 - One big bottle of broth
 - Two small bottles of LB agar
- Bought petri dishes from lab store

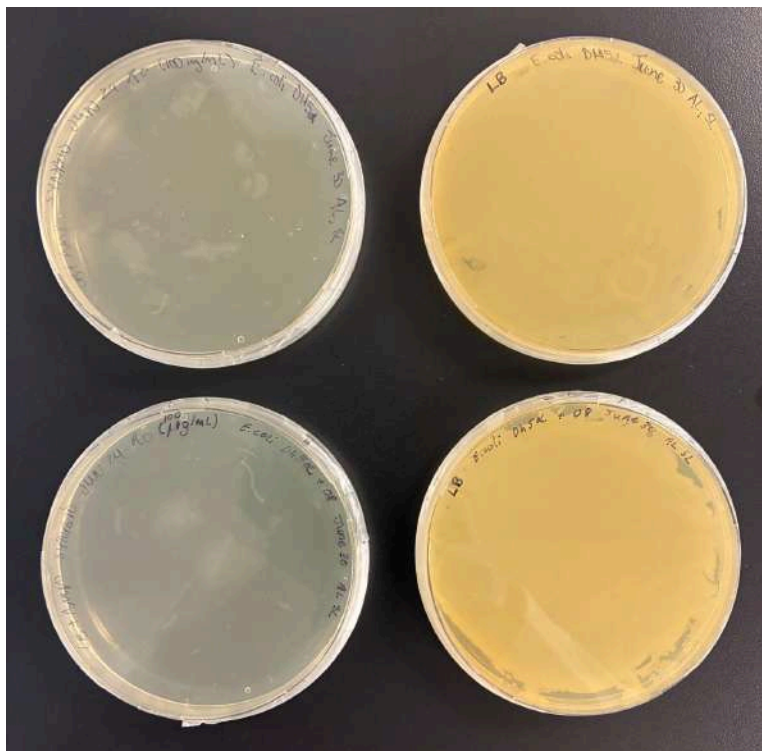


Figure 1. Transformed/untransformed plates of O8 (random GPP synthase with ampicillin resistance). We saw growth on two of our plates without Ampicillin, no growth was observed with ampicillin. Even with the transformed bacteria

Next Steps

- Troubleshooting distribution kit transformation
- Aliquotting LB media (not agar pls)
 - Eg into the 50 mL falcon tubes, have 2 at a time
- Tell people where the travel containers are kept
- Buy tape from lab store (or dollarama? good tape may only be at the lab store)

Monday, July 6

Summary

- Bought tape from the lab store and left it in CL2
- Autoclaved 2 250 ml LB agar, 1 500 ml LB broth, 2 empty 500 ml glass bottles
- Aliquoted LB media into 4 Falcon tubes
 - Stored in 4 fridge
- SOC media is about 11 ml

Next Steps

- Make plates
- Troubleshoot distribution kits

Notes

Autoclaved Items

- 2 250 ml LB agar, 1 500 ml LB broth
 - Stored in 4°C fridge
- 2 empty 500 ml glass bottles
 - Stored on cl2 shelf



Figure 1. LB broth (top shelf) and LB agars (bottom shelf) in 4°C fridge (left) and autoclaved bottles on cl2 shelf (right)

Aliquot LB media

- Aliquotted LB broth in 4 50 ml falcon tubes
- Stored in 4°C fridge



Figure 2. LB broth in falcon tubes in 4 fridge.

SOC media

- Counted in 4 fridge
- 2 vials, about 10 ml and 1 ml respectively



Figure 3. SOC media 4 fridge.

Tuesday, July 7th

Summary

- Camila notes
- 4 transformations with differing conditions
 - Transformation 1
 - Amp conc (50ug/mL)
 - Chromoprotein from 2025 kit (M5 kit 2)
 - More DNA for transformation (3 uL)
 - Transformation 2
 - Amp conc (100ug/mL)
 - Chromoprotein from 2025 kit (K23 kit 2)
 - More DNA for transformation (3 uL)
 - Transformation 3
 - Amp conc (100ug/mL)
 - Chromoprotein from 2024 kit (M5 kit 1)
 - Standard DNA for transformation (1uL)
 - Transformation 4
 - Amp conc (100ug/mL)
 - Chromoprotein from 2024 kit (M5 kit 1)
 - More DNA for transformation (3 uL)
- To reduce amt of plates, running LB controls for each transformation on one plate (split plate in half and have two streaks)
- Plates
 - 6x 100ug/mL amp plates
 - 2x 50 ug/mL amp plates
 - 4x LB plates
- Concerned abt NF water (just use milliQ)

Next Steps

- Look for growth

Notes

What was done today:

- Transformed parts into competent cells

- Did heat shock for 45sec
- Plated on LB plates for no plasmid, and parts
- Plate on LB + AMP plates for no plasmid and parts
- Made extra 2 lb + amp (100ug/ml plates)
- Did everything with filter tips

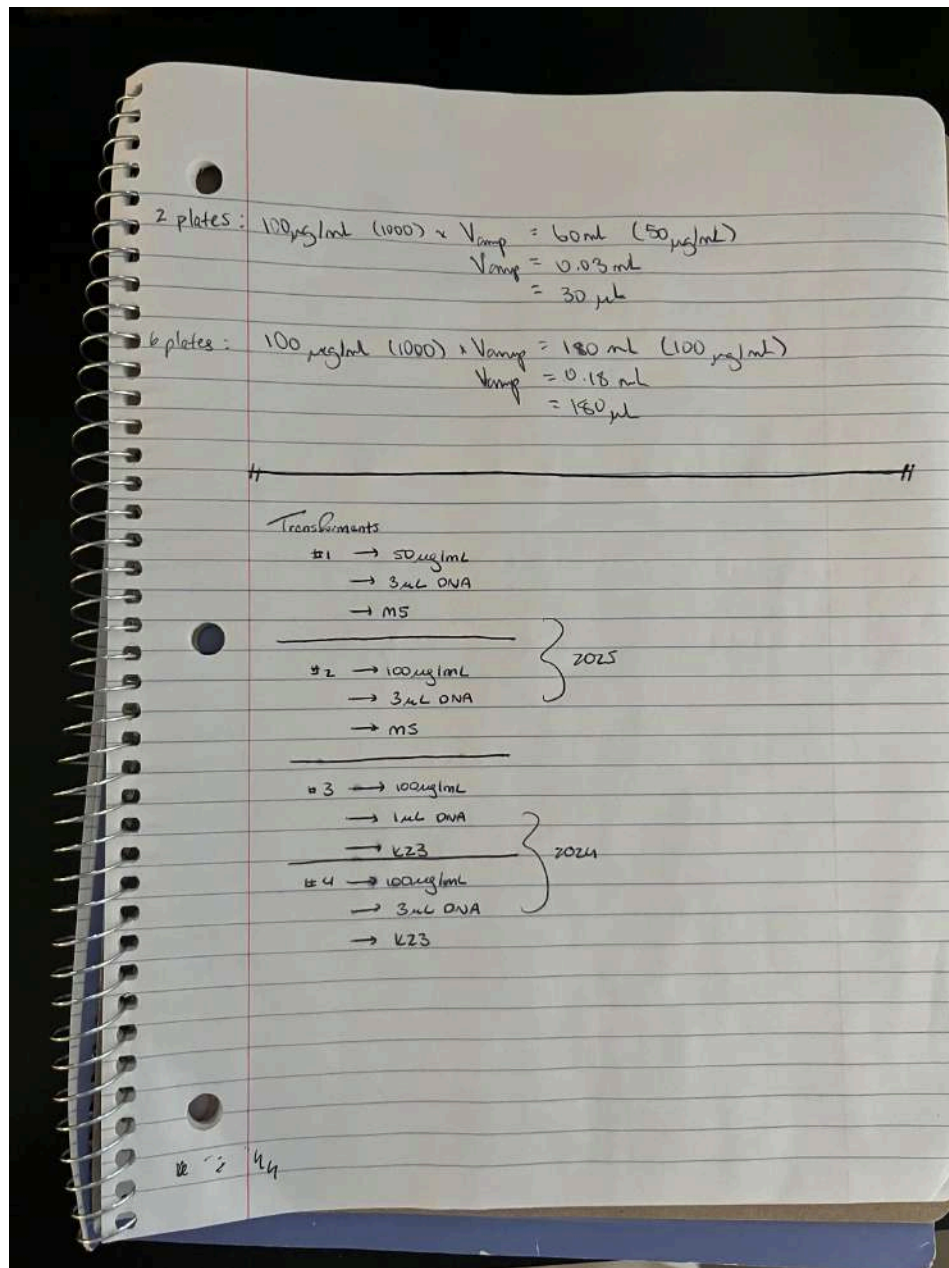


Figure 1: Calculations for transformation

Wednesday, July 8th

Summary

- 1 successful transformation (was supposed to be a pink colour, has no colour, hoping everything is fine because colour takes a while to show up) (T4, K23 2024 was successful)
- liquid passage with Liquid LB broth with amp. Made on jun 15– 215mL falcon tubes used and placed in inoculator
- Leaving remaining plates in the incubator in case
- Vaish dropped off by 4741 plasmid (which is inoculated with mix in it)
- Plasmid stored in -20
- Inoculation stored in 4
- Yeast arrived! Dropped off from Amber and placed in 4C fridge
- Dna extraction kit placed in left cabinet in CL 1

Next Steps

- Miniprep and analyze results
- Passage yeast once reagents arrive

Thursday, July 9th

Summary

- Miniprep inoculation
 - Experiment 4 (K23 from 2024 kit 2)
- Take pictures of plates (hopefully more growth)
- Quantify the amount of DNA (not today cause no plate reader training, ask Rachel tomorrow? Is it even worth it to quantify?)
- Properly label things that came in



Figure 1. Transformation plates from this week, controls all look good, unsuccessful transformations for experiment 1-3. Experiment 4 had some growth (Rachel has picture but removed plate), liquid inoculations made for T4.

Next Steps

- Transformations

Monday, July 13

Summary

- We need more agar powder
- Transformation completed by Rachel
 - Changes to protocol:
 - 200uL CC → 50 uL (excess back in -80 labelled with pink tape)
 - Heat shock at 42C for 30s / 45s / 60s
 - Used K23, 2uL of plasmid from -20
 - Poured 7 LB plates (used 3, 4 stored in 4C fridge)
 - Used 3 100ug/mL AMP + LB plates
 - Used microcentrifuge tubes, not cryotubes for heat shock
 - Spun at 3000RPM for 1 min after recovery period (dropped 45s sample oops but should be ok)
 - Placed in incubator for overnight incubation
- Made and autoclaved 3 250 ml LB agar
 - Stored in 4 fridge
- Ampicillin stock is still available.

Next Steps

- Check if successful transformation
- We need more agar powder
- Consider cutting up more parafilm

Notes

Autoclaved Items

- 3 250 ml LB agar
 - Stored in 4°C fridge
- Based on: *Solid LB Media Preparation*  Solid LB Media Preparation



Figure 1. Autoclaved LB agar in 4 fridge

E. Coli Transformation

- Done by Rachel
- Based on: *E.Coli Heat Shock Transformation* [E. Coli Heat Shock Transformation](#)
 - Used 50 ul instead of 200 ul

Footnotes

- Need more agar powder

Tuesday, July 14

Summary

- Camila notes
 - Check Rachel's transformations!
 - Report results on lab notebook(pls take good pics!! Pla make sure you can read the plates)
 - Make inoculations of successful transformants
 - Make plates (not sure if enough plates are made T-T more details below)
 - Transformations (4 total)
 - Same protocol, attention on the cryotubes and using 3uL per transformation
 - AConL (kit 2, I12)
 - mScarlet3 (kit 1, N23)
 - tENO1 (kit 2, A12)
 - AConR (kit 2, A22)
 - Controls
 - 1 for cc untransformed on amp
 - 1 for each transformation on regular LB (similar to how you did it last time streaking – ½ transformed ½ normal cc)
 - Total plates = 9
 - 5 amp plates
 - 4 regular plates

Next Steps

- please make more lb agar we are almost out

Notes

- made 4 amp plates and 4 plates
- did the transformations (3ul to 200ul of cc, 45sec heat shock)

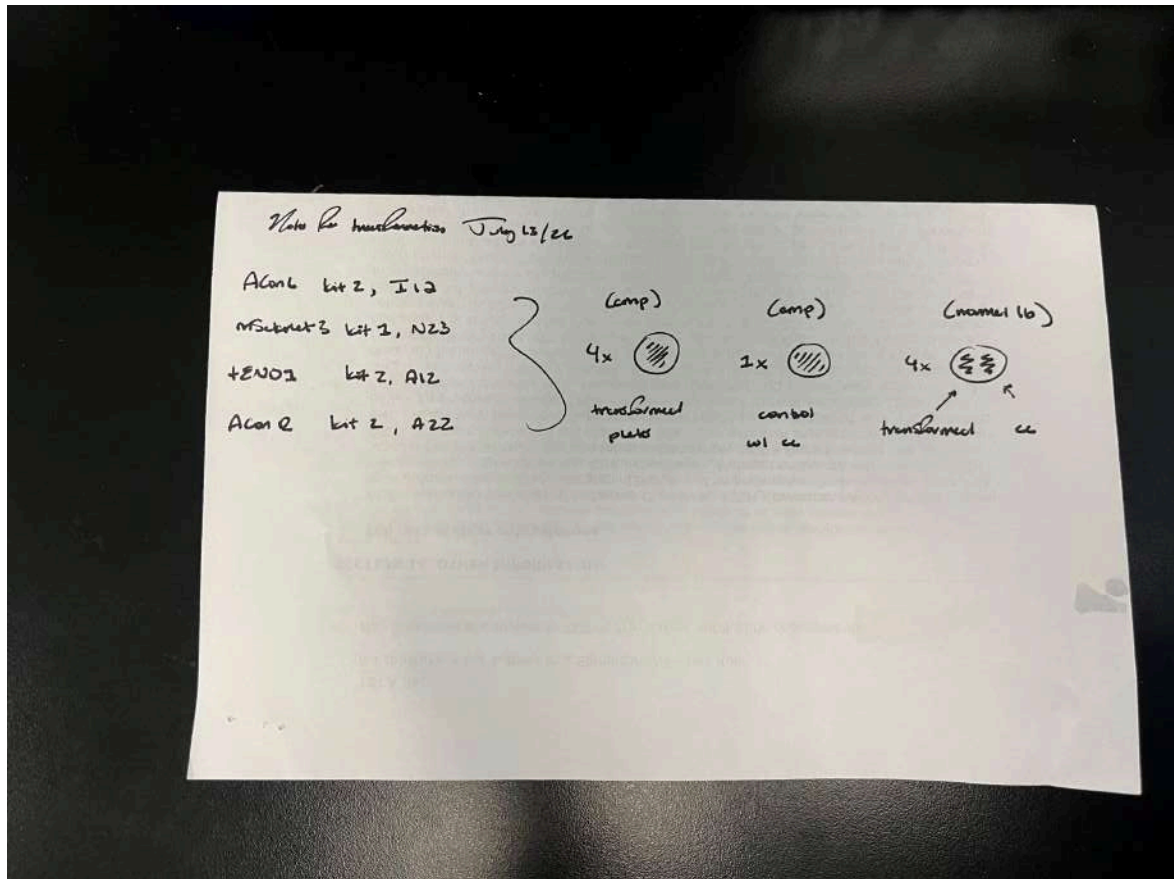


Figure 1. Notes for transformation.

- made 2 inoculations of K23 60sec heat shock from yesterday's trouble shooting
- from yesterday trouble shooting only 60 sec one worked, others were air bubbles :

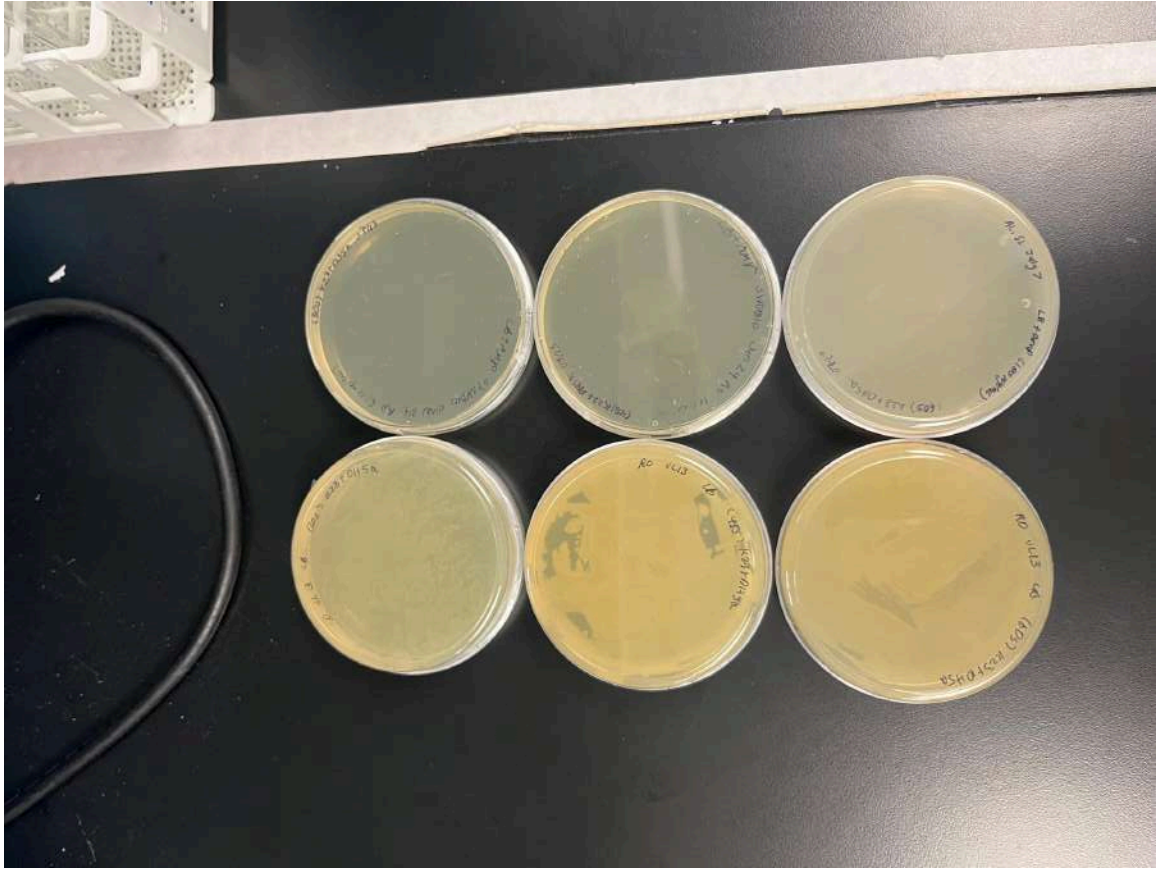


Figure 2. Petri dishes

- bought 20 petri dishes and are in cl2
- for the control, we took glycerol stock and streaked

Wednesday, July 15

Summary

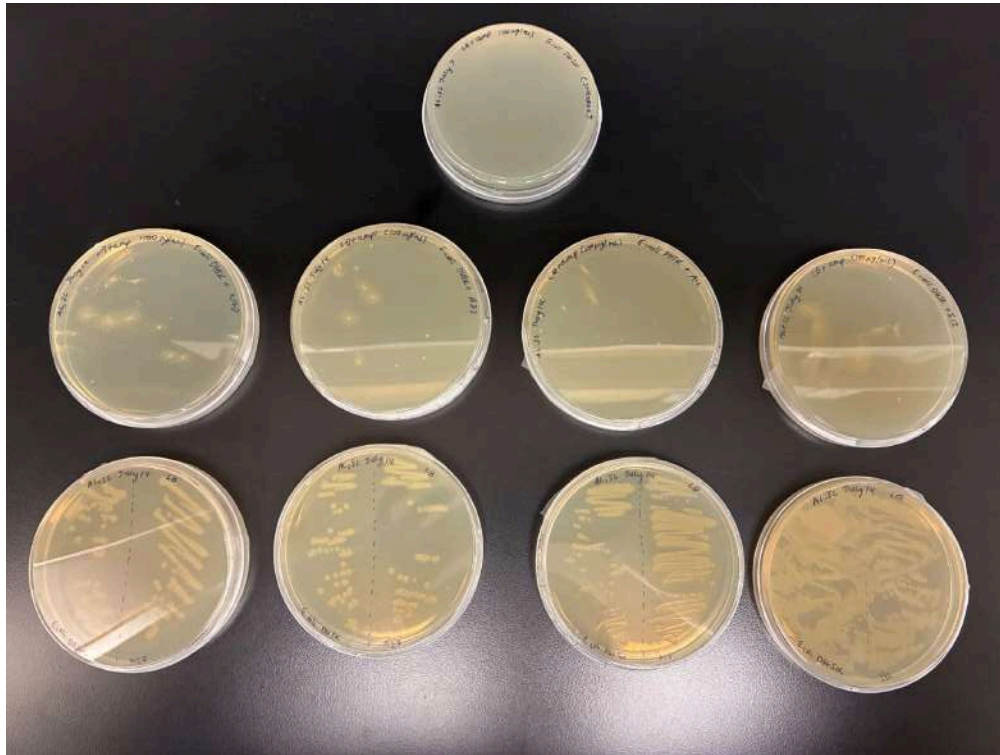


Figure 1. Petri dishes

Next Steps

- Miniprep + analyze results, start liquid culture of yeast

Notes

- Centrifuged at 4700 for 1min x2
 - Pellets were not visible
 - For step 3 of protocol
- Left in centrifuge for 15 extra minutes because it was stuck
 - Step 6
- Left in for 45sec
 - Step 7

6 Liquid cultures made

- Only 37.5ml of broth left, dispensed about 4.5ml in each
- Lid dropped on ground and ethanolated
 - For I12 (2)
- Liquid inoculation of all 4 plates, 2 each
 - Incubated

Dry Ice stored in fumehood in cl2 waiting for it to evaporate

Thursday, July 16

Summary

- Miniprepmed all samples
 - All transformations were successful, two inoculations were made from each plate
 - There are 8 miniprepmed samples ready to be quantified
- Both me (Camila) and Anthony got plate reader training and nanoquant training!
- Made inventory sheet for Emma and Jacob tomorrow so they can help with misc things if Friday ends up being buffer day we dont need

Next Steps

- Didn't have time to quantify the DNA – to do on Monday or next time Camila is in (we need to know the concentration so we know how much to run on gel tomorrow ;-;)

Friday, July 17

Summary

- Transform yeast with simple plasmid.

Monday, July 20

Summary

- Got 1 kg of agar powder from lab store
- Attempted to make gel electrophoresis
- Made and autoclaved 2 250 ml LB agar
 - Stored in 4 fridge.

Next Steps

- Prepare electrophoresis gel.
- Run gel with miniprep DNA

Notes

Autoclaved Items

- 2 250 ml LB agar
 - Stored in 4°C fridge
- Based on: *Solid LB Media Preparation*  Solid LB Media Preparation

Attempted making gel for gel electrophoresis

- Based on: *Gel Electrophoresis*  Gel Electrophoresis
- Ran into 2 issues:
 - 1. Needed to make higher volume to fill gel tray (about 140 ml)
 - 2. Could not find proper dye strain to vortex and add before casting

Footnotes

- Need to know which dye for electrophoresis in -20 fridge.
- Marta mentioned we should try to organize our plates and sections more because more people are coming to do culturing.
 - Can get extra containers from her.

Tuesday , July 21

Summary

- Camila notes
 - Passage yeast like Rachel requested :)) (we should have protocols with how much liquid media etc, lowkey we need to autoclave it tho ugh)
 - Rachel will come in to autoclave (make YNB media + YNB media agar first thing so its ready for her to autoclave?)
 - Make yeast plates with autoclaved media + agar (lets say 4 for now? Its nice to have them fresh)
 - Make normal plates + plates with amp 100 ug/ml, I can come in and do transformation in the afternoon (otherwise its too much for you guys)
 - If you want to do transformation (i can come in afternoon tho) it's part L6 on kit 1 of 2025 :))

What we did

- did transformation of L6 kit 1 2025
 - for resuspension, used NEW NF water 10ul
 - transformation was 3ul of part, 45sec heat shock, 200ul of cc
- made lb and amp plates
 - 2 extra 100ug/ml amp plates, 1 extra lb plate
- made protocol for how to make ypd media and plates using ypd broth powder
 - https://docs.google.com/document/d/1aod3w3e6ckb1sX_txjGTtXqTOGbsDKCtCtQ6Wqprl34/edit?usp=drivesdk (protocol for making media with the ypd broth powder is on the second tab)
 - we didn't make media because we don't have autoclave training
- talked with Marta about the cl2 4 degree fridge and she said we are good! she meant the first row is communal space and other people need to clean out their plates from the first row. she gave us containers to organize plates
 - if we ever need some containers to organize our microcentrifuge tubes (like those paper white boxes), there's a lot of them that we can use under the sink cabinet (of the sink closest to the cl2 fridge)

Next Steps

- make ypd media
- pls buy petri dishes! (ask for one size smaller than what we have)
- check on transformation plates

Notes

- if anyone needs extra lab coats, there's extra in cl1
 - it's in a bag, in the right most cabinet at the bottom

Wednesday, July 22

Summary

- Made 4 500mL of yeast broth
 - 2 erlenmeyer flasks, 2 500mL bottles
 - Followed protocol (halved the quantities because protocol is for 1L) x2
 - 25g ypd broth x4
 - 2 solid, using 10g agar x2
- Poured solid in 4 50m falcons, 1 15mL
- Poured 8 plates, streaked 6, 2 from each yeast plate



Figure 1. Image of 4 500 mL yeast broth.

- 2025 transformation was not successful
 - No growth on the test plate



Figure 2. E.coli petri dishes.

Next Steps

- Start prepping yeast items.

Thursday, July 23

Summary

- Checked on yeast and there is growth!
 - All the plates were moved to top incubator as it is labelled with CL2
 - Check with Marta abt that? Cause we want to keep plates in CL2 no?
- Prepared gel for gel electrophoresis
 - Tested all of our DNA ladders and loading dyes to figure out what works and what doesn't
- Cleaned up fridge and got rid of plates that were unneeded
- Passaged bacteria and made two additional plates from original stock
 - Wanted an additional set of plates from the original stock of Dr Hoare's

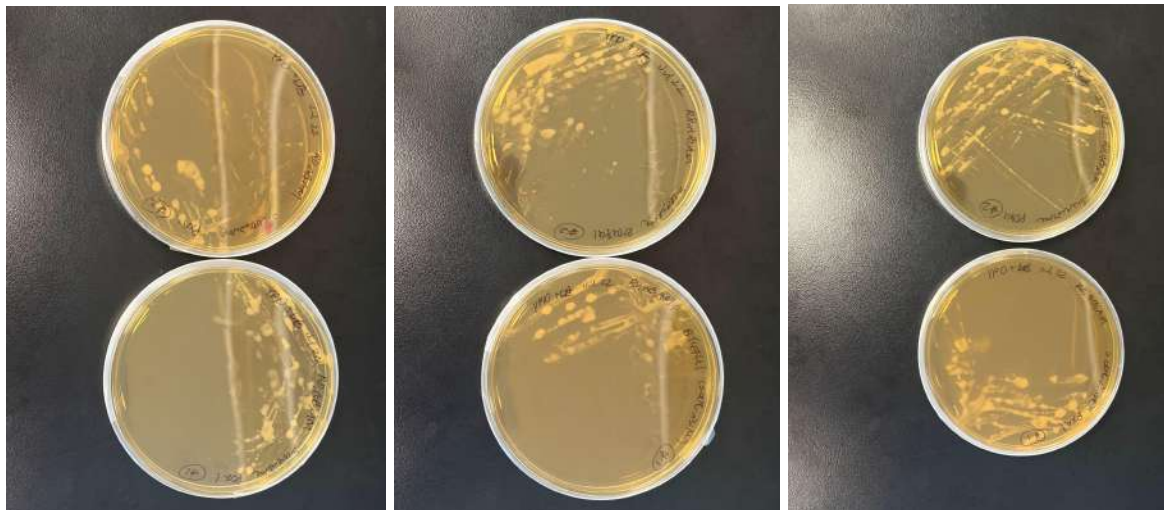


Figure 1. YPD petri dishes

Next Steps

- Nanoquant
- Put ALL plates in fridge (bacteria, yeast)

Thursday, July 23

Summary

- Camila notes
 - Inventory (mark down everything you've done!! Even if its just ensuring we have enough of smth)
 - Checked gel from yesterday (didn't look great at all, only bands were with SYMBR Green so for future put it in the gel or all the wells)
 - [10000x dilution](#) if putting in TAE buffer in the gel

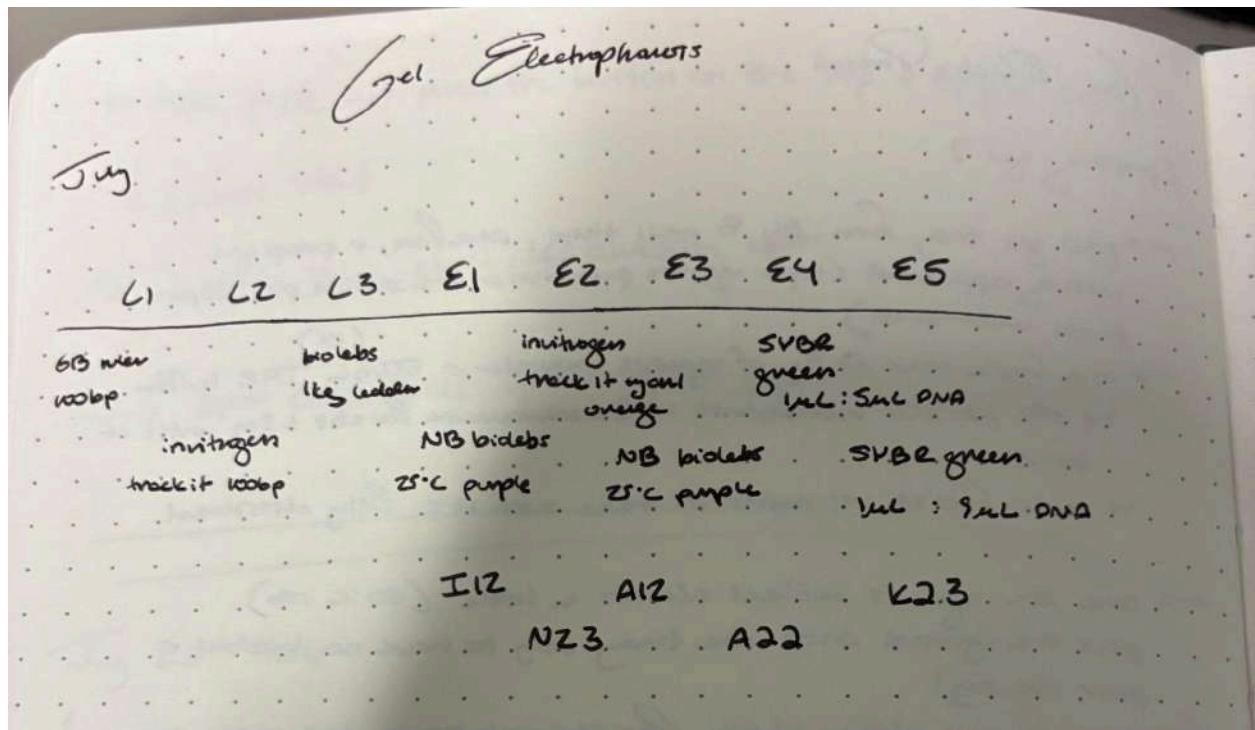


Figure 1. Gel electrophoresis layout.

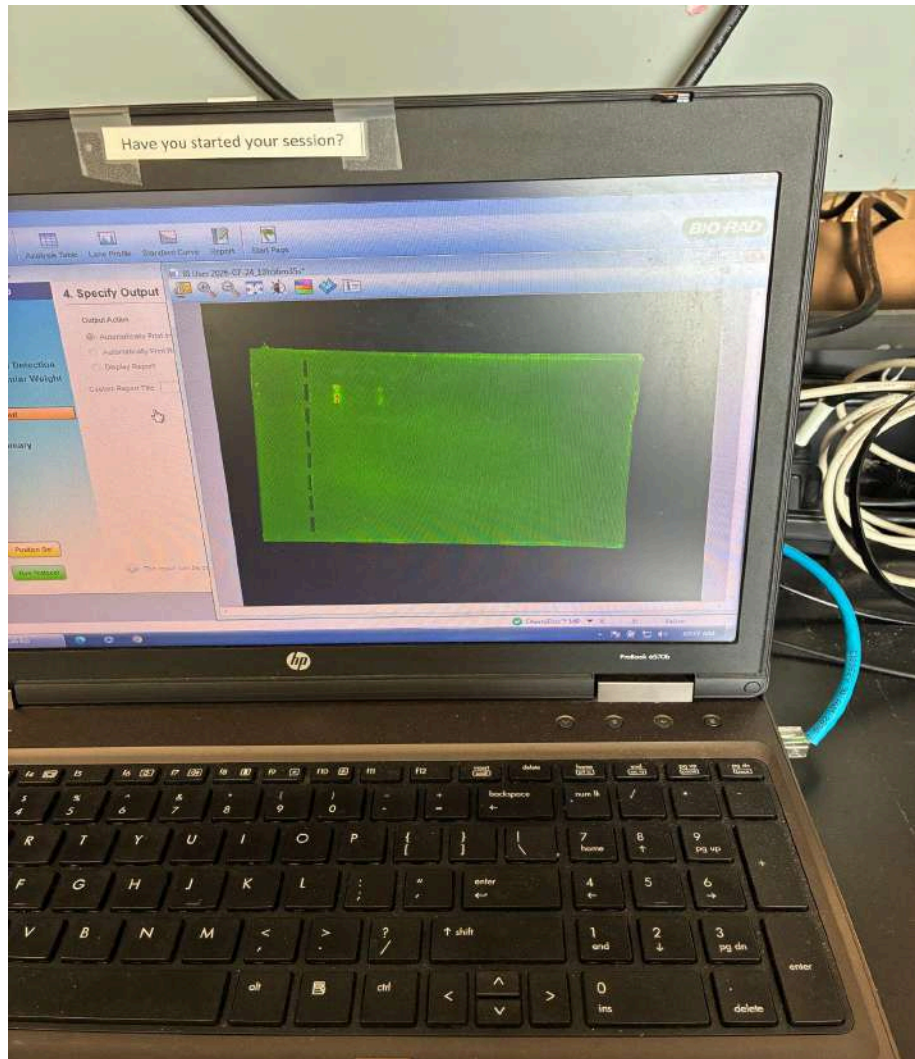


Figure 2. Gel electrophoresis reading.

- SYMBR 1uL with 5uL of DNA worked best. When diluted to 9uL of DNA and 1uL of the stain it didnt show up. For future add 5uL of the SYMBR green stain with the 50mL of TAE 1X buffer needed to make the gel

Notes

Inventory

- Made sure we have enough of each pipette tip type (filtered and normal)
- Set all pipettes to max
- Brought P20 pipette and corresponding tips to CL2
- Checked parafilm in CL2 → was good
- Centrifuge tubes in box in CL2 → NEED TO BE AUTOCLAVED

- Have more falcon tubes in CL1 underneath (in cabinet)
- **CANT FIND SEROLOGICAL PIPETTE! Couldnt find it last week, either**
- Have enough LB broth and YPD so we didn't make more
- Didn't need to buy anything → last week we bought:
 - Agar powder (100 mg)
 - P1000 + P200 tips (both filtered) → P1000 tips are a refill (have to be put in old box before use)
- Random flask in CL2 on counter... no instructions regarding what to do with it so left it (believe it can be discarded??) → PIC BELOW



Figure 3. Image of unknown flask on counter

BELOW are all the pics of the plates we currently have in the 4 degrees C fridge:

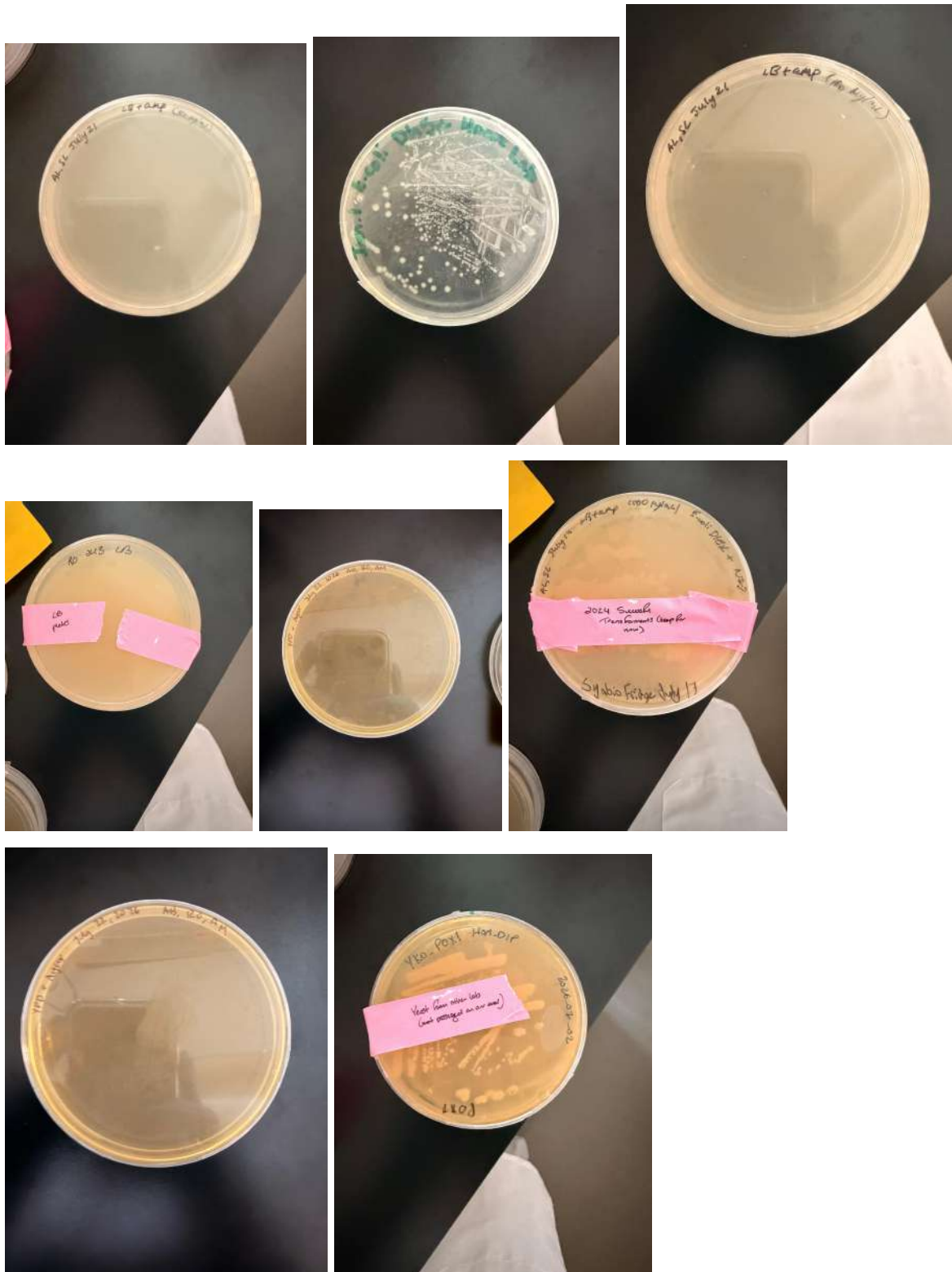


Figure 4. Petri dishes

Monday, July 27

Summary

- Made and autoclaved 2 250 ml LB agar
 - Stored in 4 fridge in c12.
- Autoclaved centrifuge tubes
 - Stored on c12 shelf
- Poured 6 plates
 - 3 plain
 - 3 LB + Amp
 - Stored all in 4 fridge in c12.
- Bought 2 10-pack Petri dishes from lab store
 - Stored on c12 shelf
- Made gel for gel electrophoresis
 - Stored in 4 fridge in c11.

Next Steps

- Transformation
- Run gel

Notes

Autoclaved Items

- Centrifuge tubes
 - Stored on c12 shelf
- 2 250 ml LB agar
 - Stored in 4°C fridge in c12
- Based on: *Solid LB Media Preparation*  *Solid LB Media Preparation*

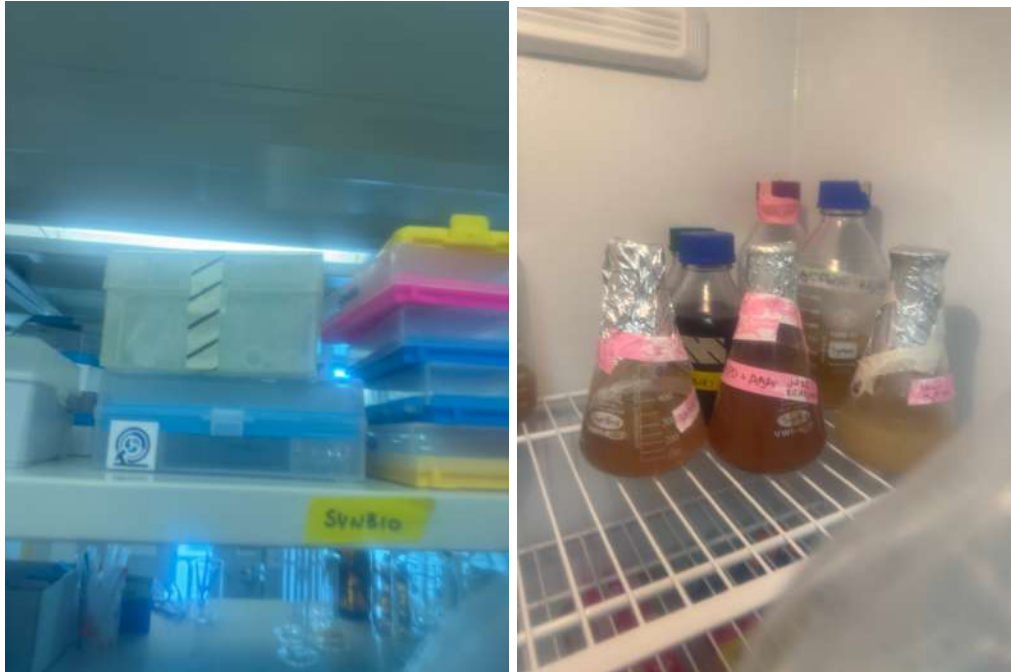


Figure 1. Centrifuge tubes on cl2 shelf (left) and LB agar in 4 fridge in cl2

Gel for gel electrophoresis

- Based on: *Gel Electrophoresis* protocol [Gel Electrophoresis](#)
 - Stored in 4 fridge in cl1 (near our bench)
 - Note: Marta said we can use this space



Figure 2. Gel stored in 4 fridge in cl1

Made plates

- Poured 6 plates
 - 3 plain
 - 3 LB + Amp
 - Stored all in 4 fridge in cl2.
- Based on: *Solid LB Media Preparation* protocol [Solid LB Media Preparation](#)



Figure 3. Plates in 4 fridge in cl2

Footnotes

- Make sure to remember to leave autoclave items $\frac{2}{3}$ the way or it will overflow. (We had some agar left behind in small autoclave in cl1 and mold).

Tuesday, July 28

Summary

- Camila notes
 - Make a couple plates for YPD agar media
 - Run gel with miniprep samples
 -
 - Ask Marta abt incubator situation
 - Can we put things in bottom incubator
 - Can we change incubator temps
- Do 1ul of stain + 5ul of sample
- 1kb ladder
- 80V
- Ran gel of miniprep samples, stain is SYBR green
 - 1ul of stain + 5ul of miniprep samples
 - 1ul stain + 3ul 1kb ladder
 - At 80V because thats the most it goes
- Gel order from left to right: ladder, I12 miniprep (attempt 1), N23 (attempt 1), A12 (attempt 1), A22 (attempt 1), K23 (attempt 1 of Camila), K23 (attempt 1 of Rachel from troubleshooting)

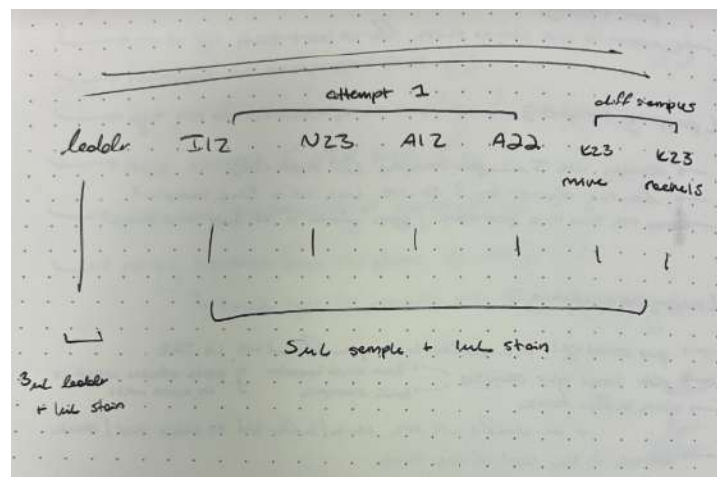


Figure 1. Layout of gel electrophoresis.

- My samples travelled the opposite direction?? samples were in the wells too



Figure 2. Image of gel in tray with power supply.

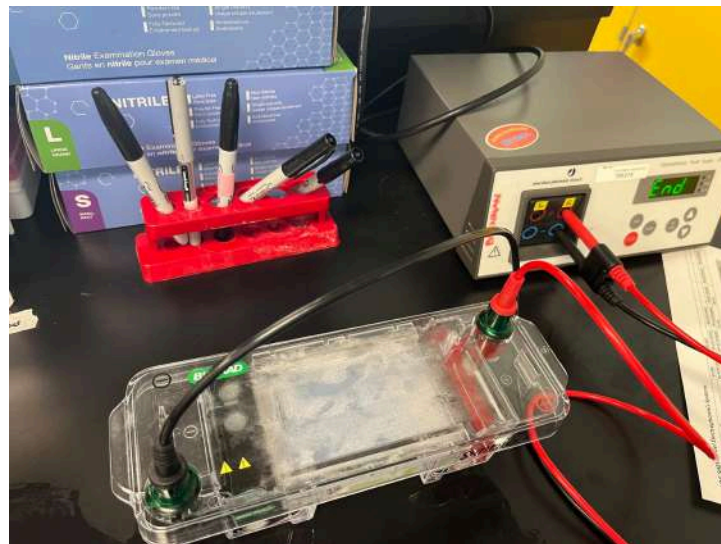


Figure 3. Setup for gel electrophoresis.

- made ypd plates, there's 12 extra in the fridge. i ran out of plates to pour so there's leftover ypd agar in the fridge
- streaked *s. cerevisiae* BY4741 on ypd plates and one is on cl2 bench at room temp, one is in incubator at 30 degree celsius
- Gel electropheris equipment is drying on cl1 bench

Next Steps

- do gel electrophoresis again
- check on plates
- buy more plates we have no more
- cut up more parafilm
- the suspicious lb agar bottle that's sitting on cl2 bench is to be thrown out. melt it, wait for it to cool so that the plastic doesn't melt, pour into a small plastic bag, throw bag in biohazard waste
- make more ypd agar almost out
- Put back dried gel electrophoresis equipment

Notes

- reminder: when autoclaving, always only autoclave half the volume of the container or else it might explode
- if looking for pipet gun, it's on cl2 top shelf to the right beside the racks. Serological pipettes are in cl2 cabinet
- do we have dry erase markers to write on the incubator whiteboard?

Footnotes

- Incubator notes:
 - Some things to note for CL2 incubators – we should NOT be putting anything in the bottom incubator labelled CL1
 - We can use the top indicator and can change the temp, write on the whiteboard of the incubator what the conditions are

Wednesday, July 29

Summary

- Camila notes

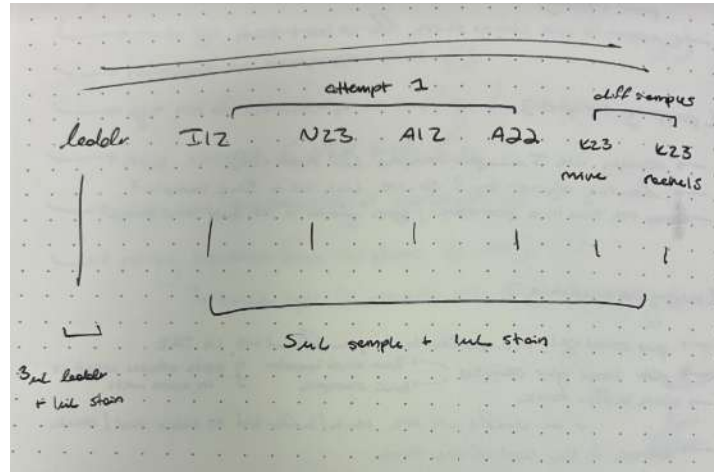


Figure 1. Gel electrophoresis setup

- This is what was done yesterday
 - Ladder is 3uL (has its own loading dye so shouldn't need anything)
 - Samples are 5uL + 1uL of loading dye
 - BE CAREFUL the loading dye isn't really heavy – have had issues with it floating before (VORTEX BEFORE ADDING PLS)
 - For our DNA use attempt 2 of everything? (I12, N23, A12, and A22 minipreps were done twice bc two inoculations were made, use the second one. K23 by me and Rachel were done twice as well, use attempt or replicate 2)
- SYBR is kept in -20 in the DNA electro white box – all parts are in the one labelled 2026 project parts box (-20)
- Loading dye that you should be using is at rt on the CL1 desktop, in the bin that's labelled 'things that don't yet have a home.' It should be in a mini rack
- Pls have a look at entry from yesterday, Andrea ran into issues where her gel was running backwards despite plugging in everything right (make sure gel is in middle and not more towards neg or positive side?)
- Pls post pics of yeast :) (pls tell me they grew)
- Casted agarose gel with 0.5g agarose + 50mL TAE + 5uL SYBR green

- Agarose gel was slightly uneven and cloudy, agarose could have been not dissolved properly despite reaching near-boiling on the hot plate and having bubbles form
 - Otherwise casting was fine
- Loaded and ran gel
 - Plugged equipment into LEFT SIDE of the power supply
 - This ran in the correct direction, but had bubbles collecting under the gel that slowly pushed towards the positive end
 - At about 25min mark, disconnected power to remove bubbles and reposition gel
 - Ran for about 40min total, was hovering between 70–75V throughout
 - Used –20C loading dye by accident (one marked as already open)
 - I12 floated out of the gel the first time, after vortexing it was fine (did two attempts total of 1uL dye + 5uL sample)
 - N23, A22, and K23 (C) floated out of the well both times

Table 1. Samples loaded into Figure 3 gel, all were replicate 2

Lane	Sample
1	1kbp DNA ladder (3uL)
2	I12 (1uL dye + 5uL sample)
3	N23 (1uL dye + 5uL sample)
4	A12 (1uL dye + 5uL sample)
5	A22 (1uL dye + 5uL sample)
6	K23 (C) (1uL dye + 5uL sample)
7	K23 (AB + RO) (1uL dye + 5uL sample)
8	Blank

- Imaged gel
 - All bands are smeared, could not excise and purify due to lack of clear bands
 - Potentially due to uneven agarose, degraded samples (please no), or SYBR green
- No growth on RT yeast, some growth on incubator yeast

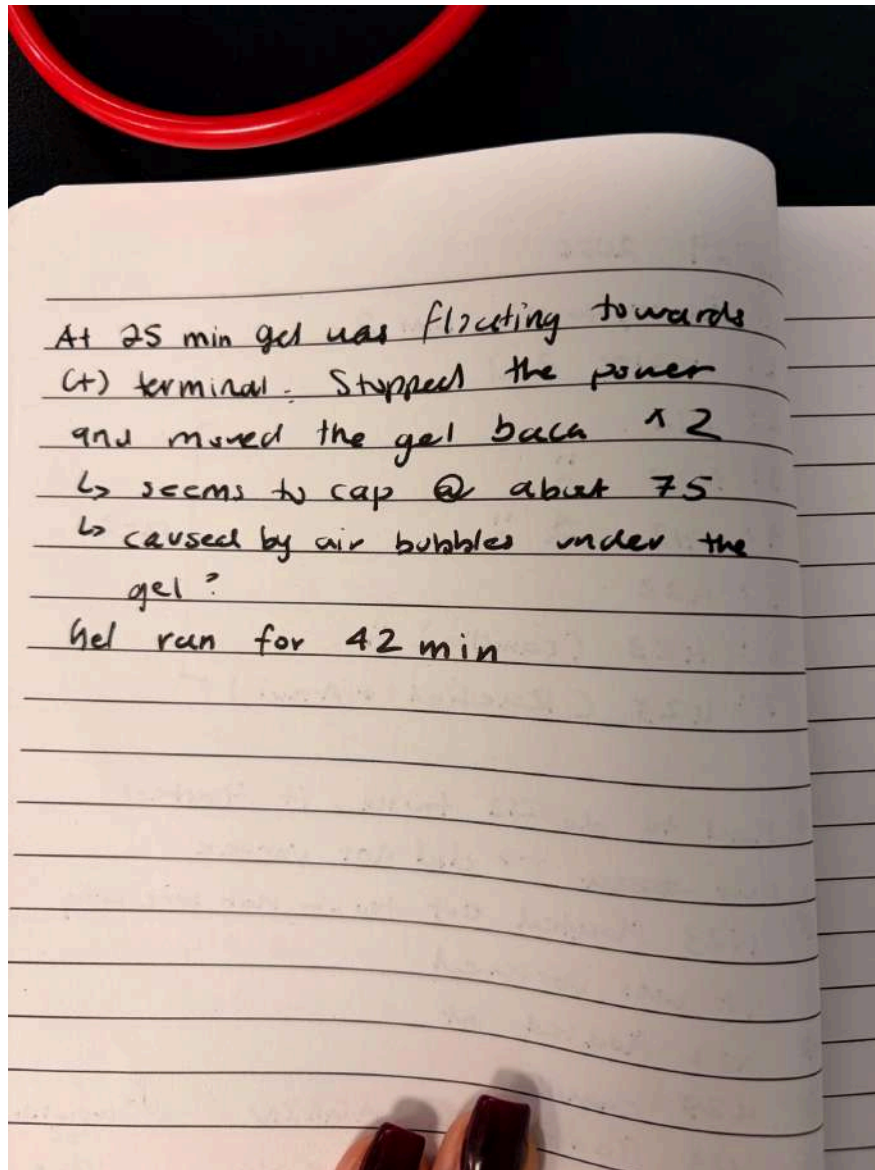


Figure 2. Notes about observation of gel electrophoresis run.



Figure 3. Setup of gel electrophoresis in CL1.

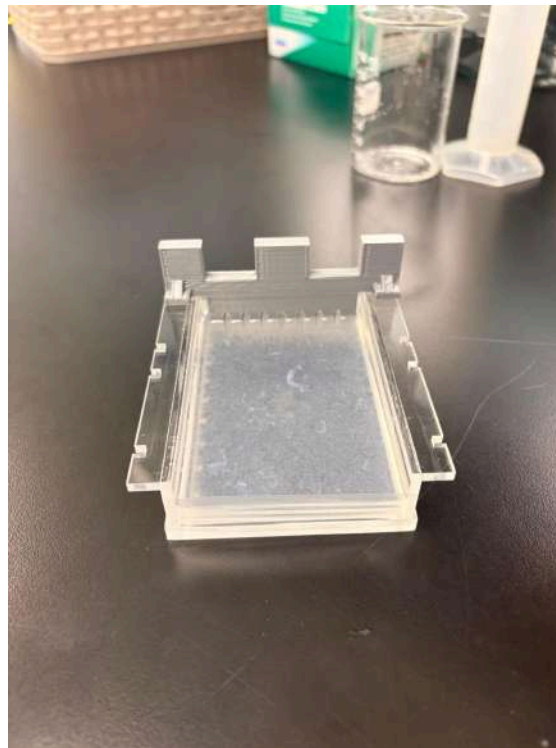


Figure 4. Image of gel in tray.

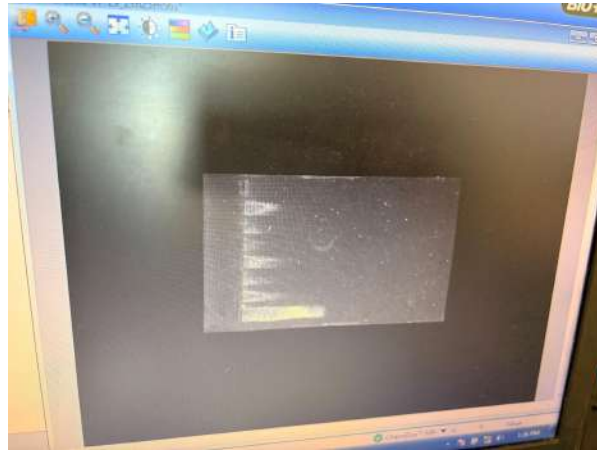


Figure 5. Gel imaging with speared bands.

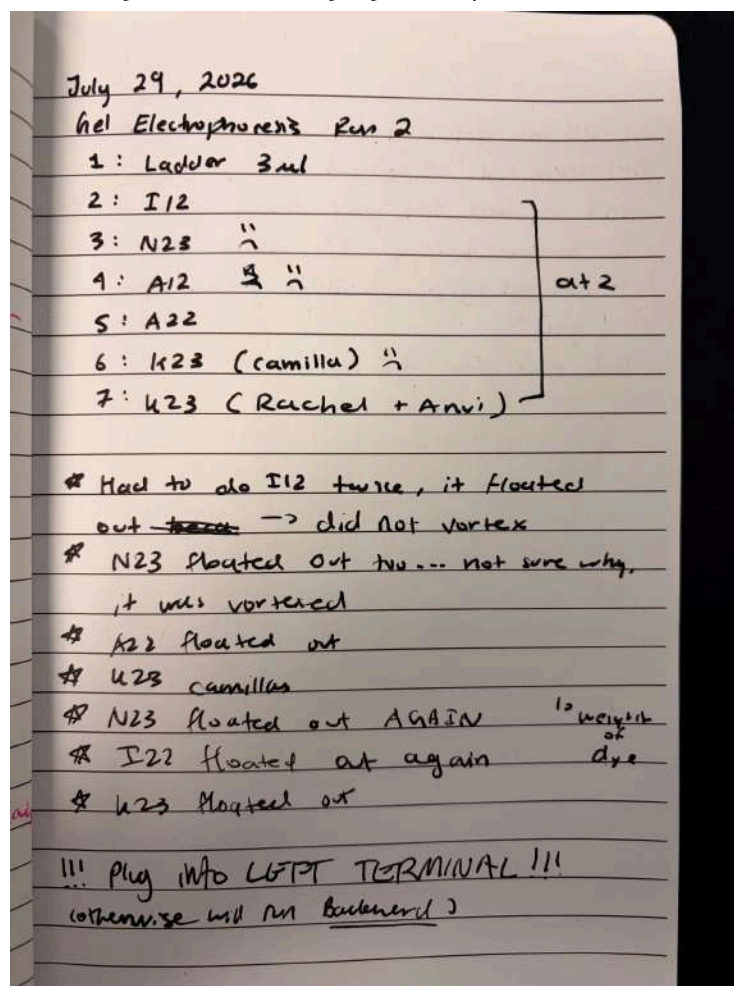


Figure 6. Notes about second gel electrophoresis run.

Next Steps

- Move -20C loading dye to RT
- Re-attempt gel electrophoresis + maybe wash gel
- Need to make more 1x TAE
- Need to come in early tmrw to make another gel

Notes

- USE LEFT SIDE OF POWER SUPPLY!

Thursday, July 30

Summary

- Gel electrophoresis was run using the same sample order as yesterday

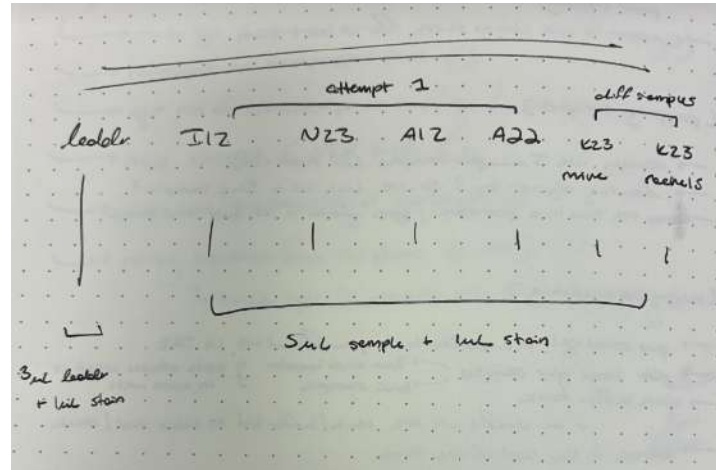


Figure 1. Gel electrophoresis layout.

- Stain wasn't put in the gel so we did a post stain
 - 5uL of SYBR green in 50mL TAE 1x (we reused TAE 1x from the gel run)
 - We put it on the shaker for 40m (the person beside ours ;-; was there the entire time in case someone came in from there lab so I could explain and beg, I figured it was harmless)

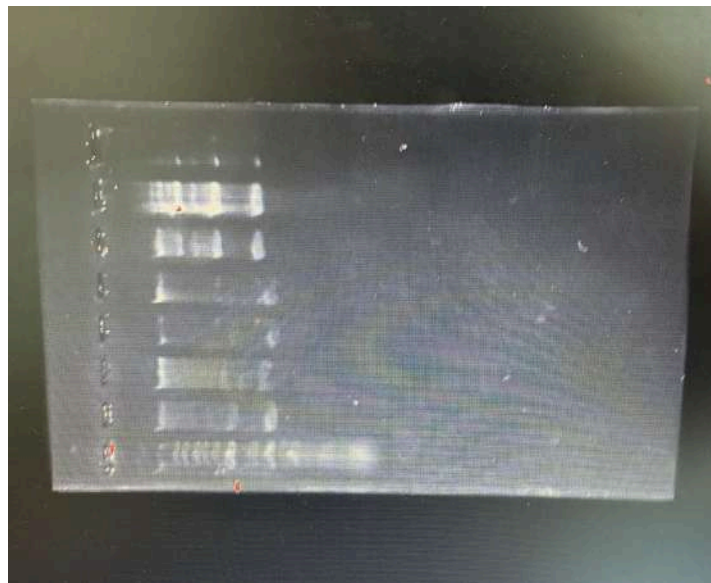


Figure 2. Gel electrophoresis visualization.

- Nanoquant values (concentrations of DNA) was evaluated in this order

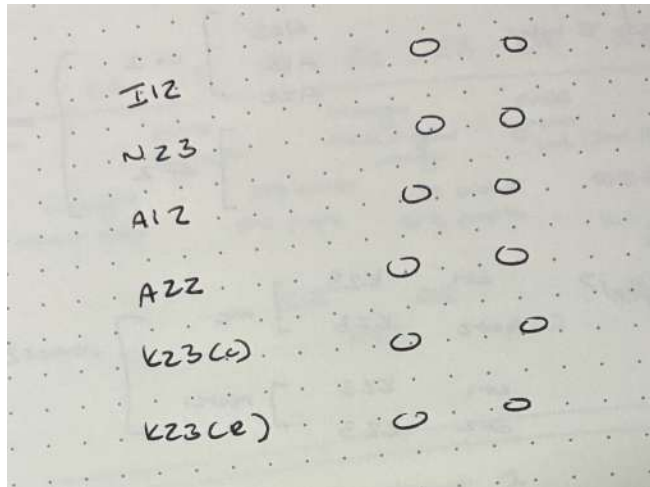


Figure 3. Nanoquant values

- Values from nanoquant

	1		2	
	Abs	Value	Abs	Value
A	260	0.0416 OD	260	0.038 OD
	280	0.0213 OD	280	0.0198 OD
B	260	0.0871 OD	260	0.0856 OD
	280	0.0444 OD	280	0.0435 OD
C	260	0.0388 OD	260	0.0807 OD
	280	0.022 OD	280	0.046 OD
D	260	0.0614 OD	260	0.0595 OD
	280	0.0321 OD	280	0.0311 OD
E	260	0.0718 OD	260	0.0688 OD
	280	0.0389 OD	280	0.0368 OD
F	260	0.0775 OD	260	0.0809 OD
	280	0.0416 OD	280	0.0435 OD
G	260	OD	260	OD
	280	OD	280	OD
H	260	OD	260	OD
	280	OD	280	OD

Figure 4. Values from nanoquant

- Yeast growth
 - More than yesterday, 30 deg looks good! More than rt so going forwards 30 deg might be best
 - Was left in incubator to be evaluated again tomorrow



Figure 5. Image of yeast growth plates at 30 C (left) and room temperature (right).

- Reagents were delivered to lab and put directly to refrigerator in 4 deg CL2
 - ssDNA
 - Uracil dropout
 - Ammonium sulfate
 - Smth i forget whoops

Next Steps

- Gel extraction of bands
- Put yeast in fridge
- Inventory
- Note down everything we received yesterday (should be in rack in CL2 4deg)

Friday, July 31

Summary

- Extracted N23 and K23 from gel from yesterday
 - Remaining gel still in CL1 fridge
 - Extracted DNA in -20C fridge in CL2
- Checked on yeast
 - No picture, but more growth in 37 degree incubator than RT bench
- Took inventory
 - Added reagents under fume hood to inventory
- Aliquoted isopropanol to 15mL tube, stored on the bench in CL1

Next Steps

- Maybe test the integrity / quantity of extracted DNA
- Prepare for when reagents arrive to do yeast transformation

Notes

Gel extraction

- Followed [Norgen's protocol](#):
 - Mass of N23 = 0.19g → 570uL of buffer G → 190uL of isopropanol
 - Mass of K23 = 0.21g → 630uL of buffer G → 210uL of isopropanol
 - Used 50uL of elution buffer
 - Used stock ethanol from BI since we couldn't find the ethanol under the fume hood at the time (oops) for buffer A
- Needed to take P1000 out of CL2 to use for this protocol
- Needed to take sample into CL2 to use water bath for this protocol (55C for 4 min)
- Extracted the brightest, nicest bands from gel (K23 and N23) using a spatula
 - DO NOT EXCISE IN CHEMIDOC, took gel out to make cuts and put it back to check

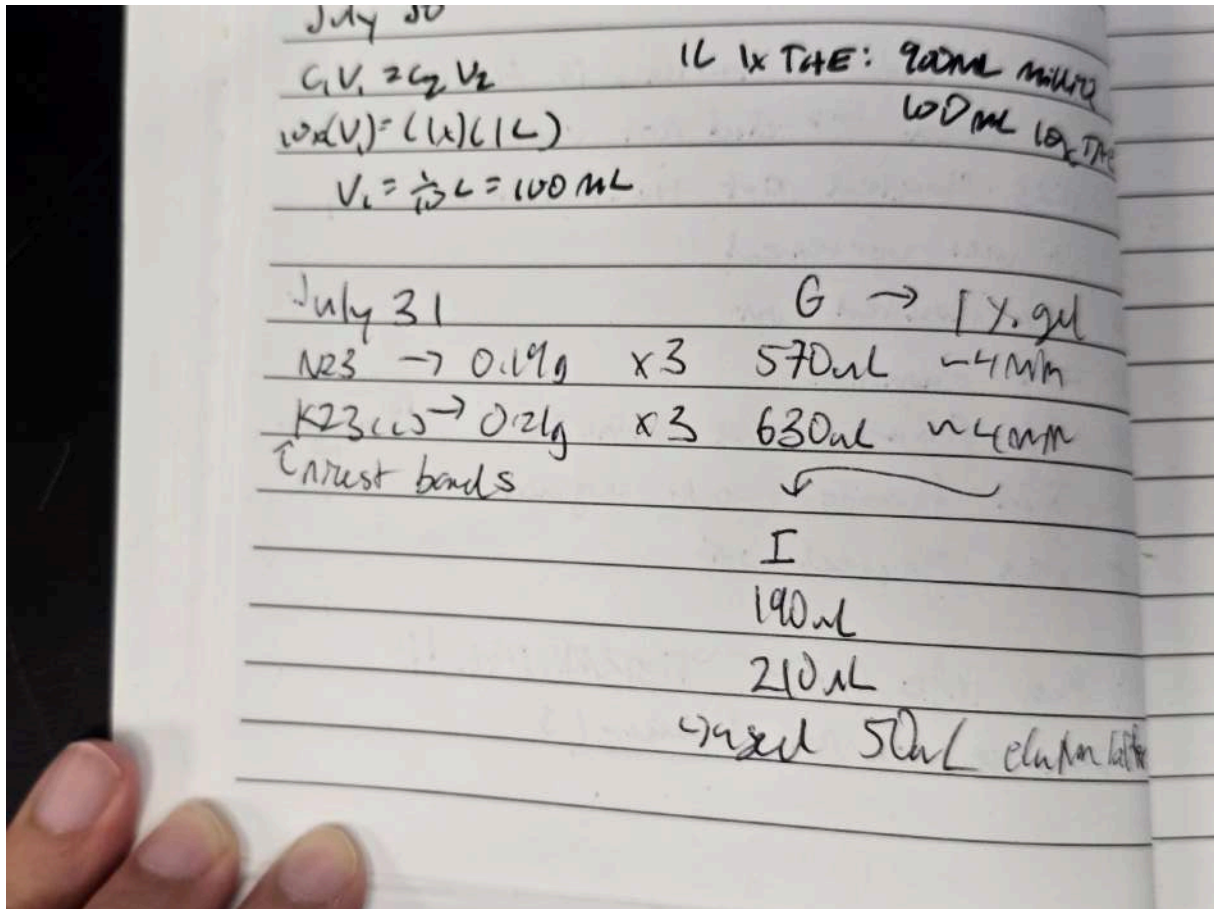


Figure 1. Calculations for gel extraction

Inventory

- Took inventory of items in CL1 and CL2, added cabinet under fume hood and its contents to inventory

Tuesday, August 4

Summary

- Camila notes
 - Transform URA3 into bacteria
 - Kit 2 O16
 - Normal 3uL into 200uL of cc at 45s heat shock
 - Liquid inoculations of transformants
 - Choose 2 colonies per transformant (8 inoculations total)
 - A12
 - A22
 - N23
 - I12
 - You'll probs need 40mL of LB + amp at LEAST, message Rachel early if you need more autoclaved media LB
 - Pls tell me Friday ppl took yeast out of incubator and rt and put in fridge ;-; (if not pls put the two plates in fridge tyy)
 - IF TIME
 - My inventory [checklist](#) 😊

Notes

- Bought 2 packs 15ml falcon tubes, 20ul + 10ul filter tips, 2 packs of plates
 - We can return the tips if the package is unopened
 - P10 filter tip box is in the right cupboard in cl1
- Made inoculations for A12, A22, N23, I12
 - Did 5ul LB + 5ul of 100mg/ml AMP
 - Each part has 2 replicates, one for each colony
 - Currently loosely capped and shaking in cl2 incubator
- Yeast plates were not taken from incubator and rt to be kept in the fridge on Friday, so i put it in the fridge



Figure 1. Yeast YPD plates.

- Did transformation of O16, 2024 kit 2 part
 - 3ul of plasmid, 200ul cc, 45sec heat shock
 - Plates for transformation
 - E.coli + O16 on LB + 100ug/ml AMP plate, E.coli + O16 on LB plate
 - No treatment E.coli cells on LB + 100ug/ml AMP plate, No treatment E.coli cells on LB plate
- cut parafilm and did most of camila's checklist
- found blue whiteboard marker! it's in cl2 basket with sharpie

Next Steps

- Make more lb pls and lb agar? We only have 3 lb plates and 3 lb amp plates left

Wednesday, August 5

Summary

- Attempted to miniprep liquid cultures from batch 2 (see below for detailed notes + results)
 - Reached until step 2 of the protocol
 - The mini centrifuge tubes were put in the incubator until we decide what to do with them– hopefully we see growth, may have to be discarded
- Made liquid cultures
 - Only one colony was present on the plate
 - A small blotchy colony was also picked from the side of the plate
 - 2 made cultures made

Next Steps

- Let the bacteria incubate and grow for longer before attempting the miniprep again

Thursday, August 6

Summary

- Make bacteria cc + miniprep URA3

Next Steps

- Buffer day

Friday, August 7

Summary

- Mostly an inventory day
 - Reset all the pipettes to their max volume
 - Had enough filtered and unfiltered tips → didn't need to buy any more
 - Good on parafilm in CL2
 - Still had enough autoclaved centrifuge tubes from last time we checked
 - Serological pipette charged to 4
 - Had enough falcon tubes
 - Got rid of LB agar that had been sitting on the bench in CL2 for awhile (heated, dumped in bag → triple bagged; disposed of)
 - Didn't need to buy any more plates (looks like someone had just bought a new bag → however, I did use a few, so there are only ~6 left I think)
- Made 4 LB + amp. Plates in BSC and refrigerated → already had 3 LB and amp. Plates in fridge and 4 LB
- Made more LB + agar stock that needs to be autoclaved
 - Used 250 mL from pre-made LB broth stock in fridge and 3.75 g agar

Next Steps

- Plates from August 6th (or 5th??) did not show any growth, unfortunately :(
- Camila wants to know more about how to make YPD media without uracil → didn't end up making any but probably will need to soon
 - Didn't see that it was in a protocol when I first checked but I missed it → *Liquid SD Media Prep.* → there are instructions on how to do this

Notes

- The protocol above details making SD without uracil, not YPD
 - The protocol we have isn't for YPD without uracil, it's for **SD-Ura.**
 - If we need uracil selection, use SD-Ura rather than trying to make YPD without uracil.

Monday, August 10

Summary

- Camila notes
 - Make selective yeast plates / selective yeast media
 - 400 mL of media
 - 400 mL of agar
 - Yeast inoculation (make at least 2, shaking at 200rpm) – using general YPD
 - Make more YPD? I think we have quite a bit?
 - Make lithium acetate solution (0.1 M solution and 1 M solution) in water, use heat if needed but should be fairly soluble
 - Filter sterilize
 - Can be done outside of bsc (make sure area is ethanolized)
 - Might be helpful to watch video for best practice (in terms of keeping everything sterile)
 - Prep filter (yellow filters, found in CL1 in cabinets under bench space)
 - Grab syringe (I think we have 10 mL or more? Grab however much needed to make 40 mL)
 - In CL2 screw the filter onto the syringe, pop the top of the syringe off, pour in the LiOAc soln, compress solution into falcon tube by using syringe plunger
 - Make PEG 3500 50% w/v

Place 50 g of polyethylene glycol, MW 3350 (Sigma) in a 150 mL glass beaker and add 35 mL of ddH₂O. Stir with a magnetic stirring bar until dissolved. This will take about 30 min. Transfer all of the liquid to a 100 mL graduated cylinder. Rinse the beaker with a small amount of distilled water, add this to the graduated cylinder containing the PEG solution, and bring the volume to exactly 100 mLs. Mix well by inversion. Filter sterilize using a 0.45 µm filter unit (Nalgene), and store in a securely capped bottle.

NOTE: YOU CAN ALSO AUTOCLAVE PEG TO STERILIZE. Its a bit easier.

*Note:

Evaporation of the water from the PEG stock solution will result in an increase in the effective concentration of PEG in the transformation reaction and severely reduce the efficiency.
 - Put ssDNA in freezer (-20) - in the little centrifuge aliquots
- What was done:
- Made 4 3mL yeast inoculations (2 from each plate) + put 10mL of YPD in incubator for spectrophotometry
- Autoclaved 400mL liquid SD (NO DEXTROSE), 400mL solid SD (NO DEXTROSE), and 100mL 50% PEG 3350 based off of protocol from the lab we got the reagents from

-
- Prepared 100mL 40% dextrose and 100mL 50% PEG 3350 for filter sterilization, however:
 - Had issues filter sterilizing dextrose (would not pass through filter + syringe exploded), as a result 400mL solid SD is likely contaminated and will probably need to remake dextrose
 - Received lithium acetate and Nile red, Nile red was put in 4C fridge and lithium acetate is on CL2 bench
 - Moved ssDNA to freezer

16

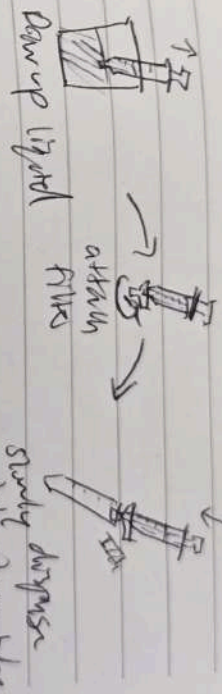
→ + 8g agar

- 300mL milk + 0.7g MVB + 2g MMS, + 0.02 x 4 (0.768g)
- Heat ~ 5min (Do NOT BOIL)
- Fill to 380mL w/ pure distilled water gradually
- Transfer to 500mL bottle + autoclave
- add 20mL 40% glucose
- filter sterilize first!
- 40g D-glucose + ~~100mL~~ 30 ~~sterile~~ conl
- add ddH₂O to 100mL in cooler

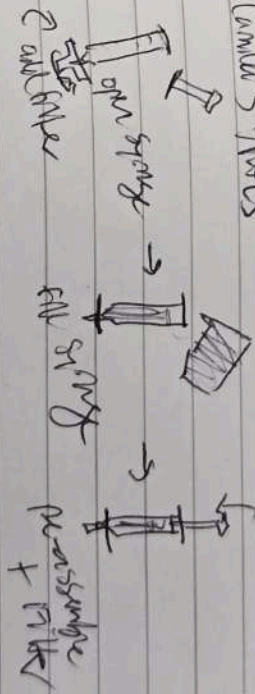
PEG

- 50g PEG 350mddH₂O, 150mL water
- stir until dissolved, ~ 30min
- Rinse and add ddH₂O until 100mL
- ↳ into grad cylinder
- ↳ mix well w/ MMS
- ↳ filter sterilize

How to filter ~~sterile~~ ^{sterilize}



Canister's Notes



17

Next Steps

- Re-attempt filter sterilization of dextrose, PEG, and lithium acetate
- Re-make SD solid media (will just put dextrose in if filter sterilization doesn't work)
- Yeast transformation

Notes

○

Tuesday, August 11

Summary

- Filter sterilized dextrose, 1M lithium acetate, 0.1M lithium acetate
 - 0.1M lithium acetate was made by 5mL 1M + 45mL water
- Transformed part O12 into yeast
 - 11ul plasmid + 14ul NF water

Handwritten calculation on a grid background:

$$(0.0464) \times = \left(\frac{0.3}{25} \right) (25)$$

$$x = 10.78 \mu\text{L DNA}$$

$$25 - x = 14.22 \mu\text{L water}$$

Annotation: O12 part B in miniprep aliquots (blue box in -20°C)

Figure 1. Calculation.

- For the SD plate with just regular yeast, we took yeast, centrifuged, washed with NF water, resuspended in NF water to plate
- Plates:
 - 2 YPD: one was regular yeast, other was yeast + O12
 - 3 SD: one was regular yeast, other was yeast + O12 (100ul), yeast + O12 (200ul)
 - Wanted to see if there is better growth with 200ul
- Boiled ssDNA and then aliquoted into MANY pcr tubes (5ul each), in the pink box in freezer -20
 - Note: they have now been boiled twice
- Camila found genes!!!!

What Rachel did:

- Autoclaved
- Took OD
- Mixed PEG
- Made plates!, also bought plates

Next Steps

- Look at plates
- Transform bacteria, PCR genes

Notes

- Marta says if we need to use a heat block, we can use the one in cll
- Do we want to return the p10 tips that are unopened and sitting on cll bench? It expires in 2027 and i bought it last week and he said i can return it if i want

Wednesday, August 12

Summary

- Camila notes
 - Check plates (take pics if any inkling of growth! Leave in incubator)
 - Transform two parts from iGem 2024 kit
 - First resuspend them from kit using following

Using the Kit

1. To resuspend the DNA, briefly centrifuge the plate (optional) and pipette 10 μ L of dH₂O into the desired wells. Pipette up and down a few times and wait for 5 minutes. Pipette up and down a few times again to fully resuspend the DNA. Do not forget to change tips from one well to another.

- Take out part from KIT 2 (its marked on the sides of plate), M6
- Use up 3 plates
 - Experiment plate (amp + transformed comp cells) – pipette sol
 - Negative plate (amp + control cells) – pipette soln
 - Positive plate (separated in half, lb + comp cells) – streak cells
- Transformation doing 3 μ L of DNA, 45s heat shock, 200 μ L of comp cells (not sure if its updated on protocol)

If time!! Please make a gel T-T tomorrow is long day cause we're doing pcr/mutagenesis, running gel, extracting gel, and pcr cleanup

- Made gel (it's in CL1 fridge)
- Resuspended M6 with NF water on bench (but it's from last year so it might be bad)
- Transformed E. coli with M6, placed in bottom incubator (37 degrees)
- Checked yeast, no growth except on YPD plates (+ control)

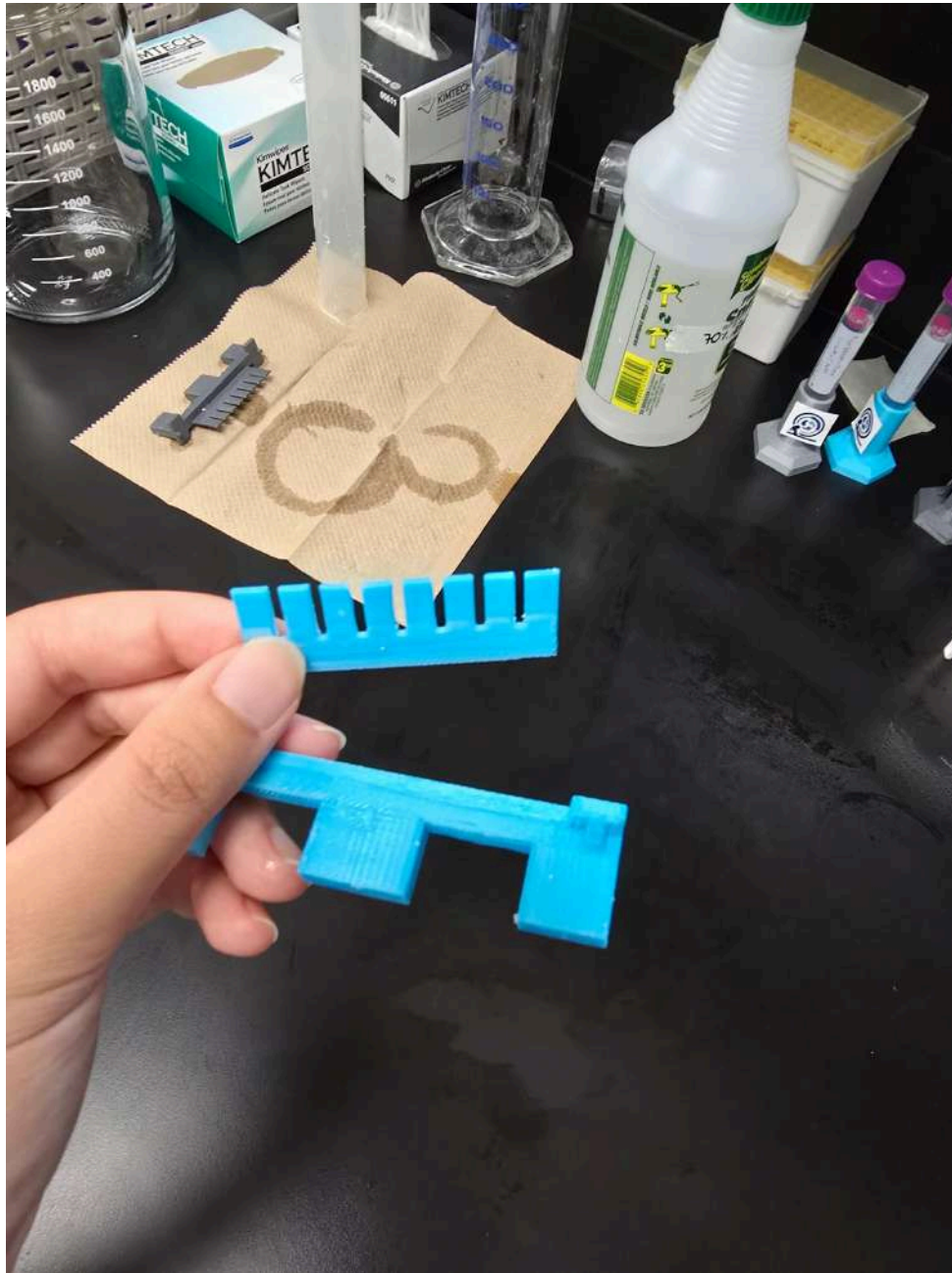


Figure 1. Broken comb for gel.



Figure 2. Gel for electrophoresis.



Figure 3. Plates.



Figure 4. Yeast plates.

Next Steps

- Liquid inoculation of transformation + check on yeast transformation

Footnotes

- One of the gel combs broke, we have a second but I'll look into 3D printing another

Thursday, August 13th

Summary

- PCR
 - RNAt 1,2,3
 - Standard pcr with amplification temp at 34 ($^{\text{TM}}$ = 37.6)
 - cpTEF1
 - Standard pcr with amplification temp at 42 ($^{\text{TM}}$ = 46.6)
- Ran gel with PCR products
 - Had to run several as people had made gels with agar instead of agarose
 - Also had to run one more time as 1kb ladder was used instead of 100bp
- The one successful gel

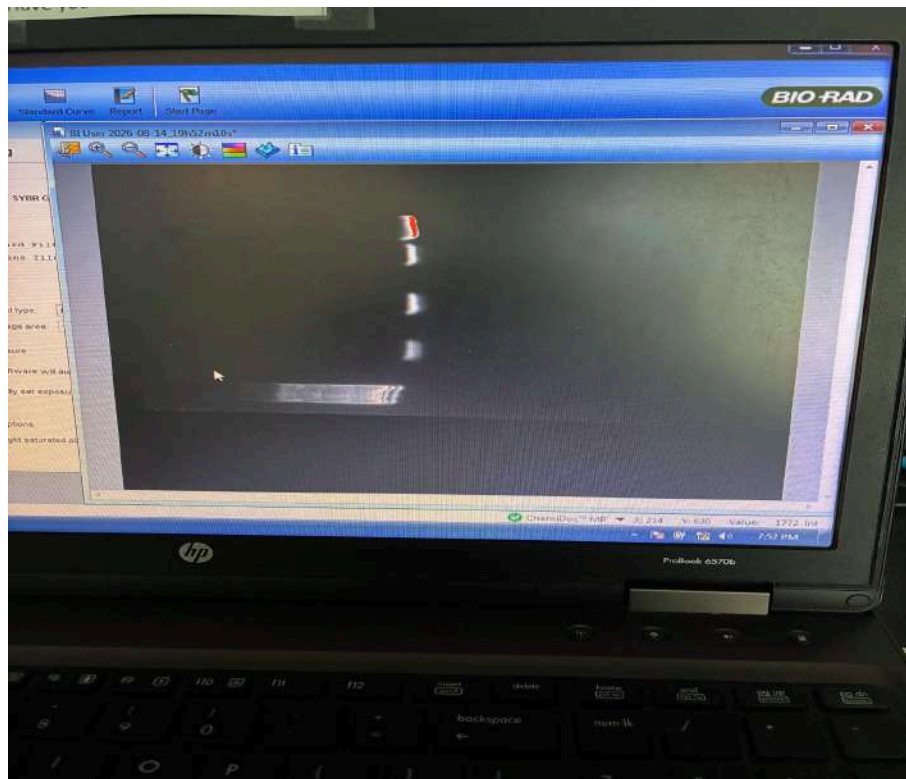


Figure 1. Gel electrophoresis analysis.

- The gels were cut and put into centrifuge tubes for gel extraction on Saturday
- Plate photos of Yeast growth (as of August 13th,2026)

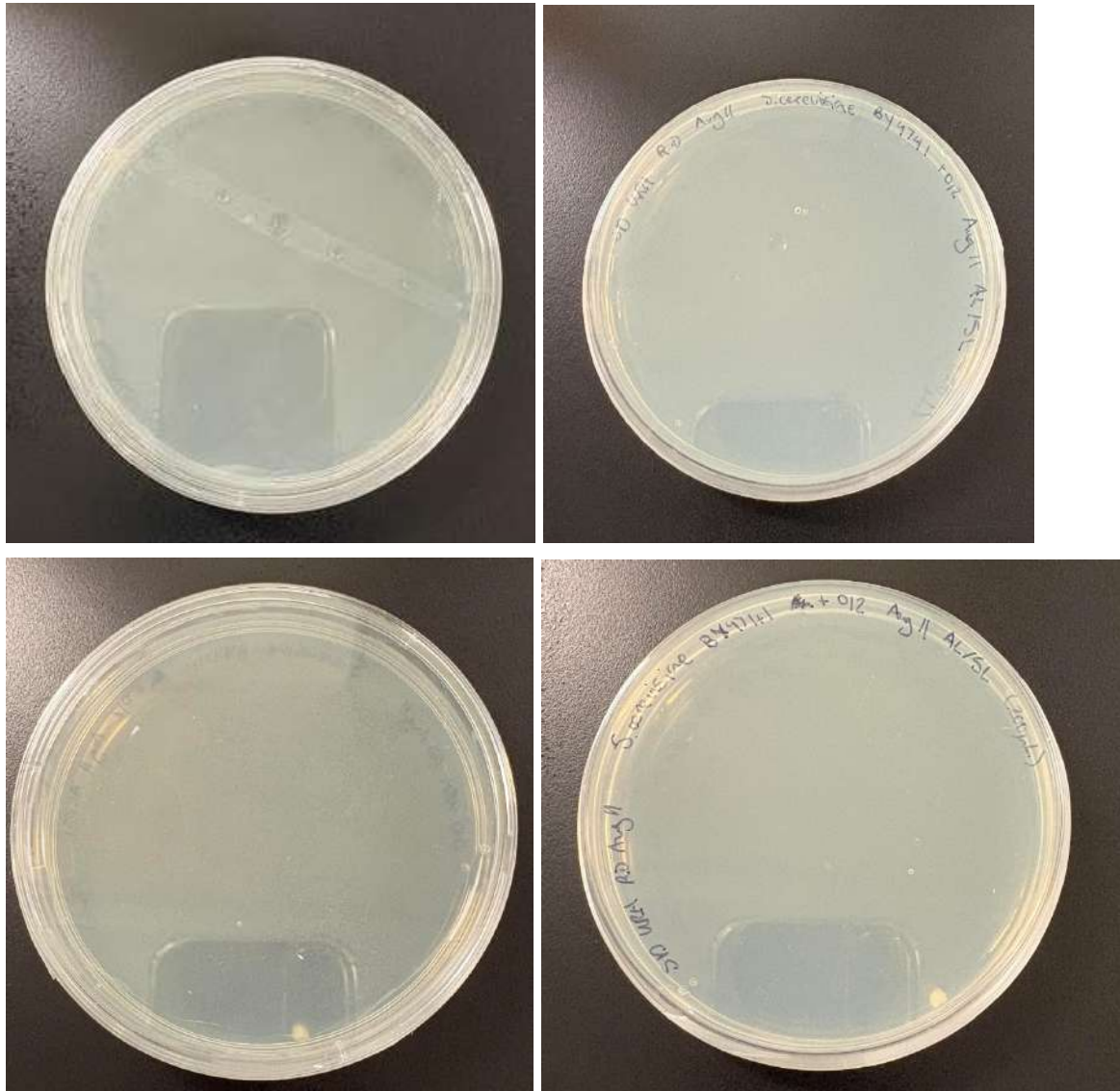


Figure 2. Plates of yeast growth.

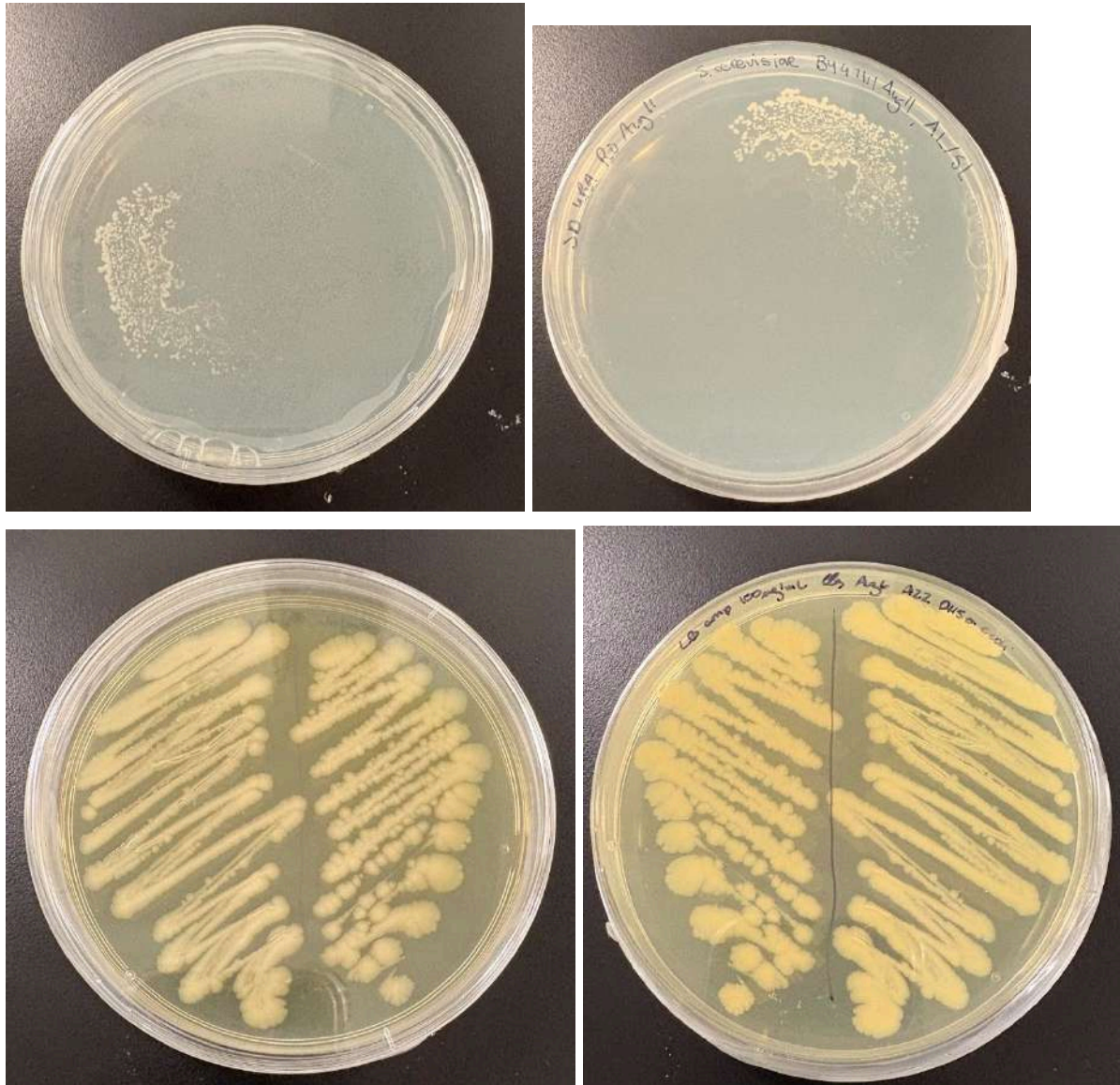


Figure 3. Plates of yeast growth.

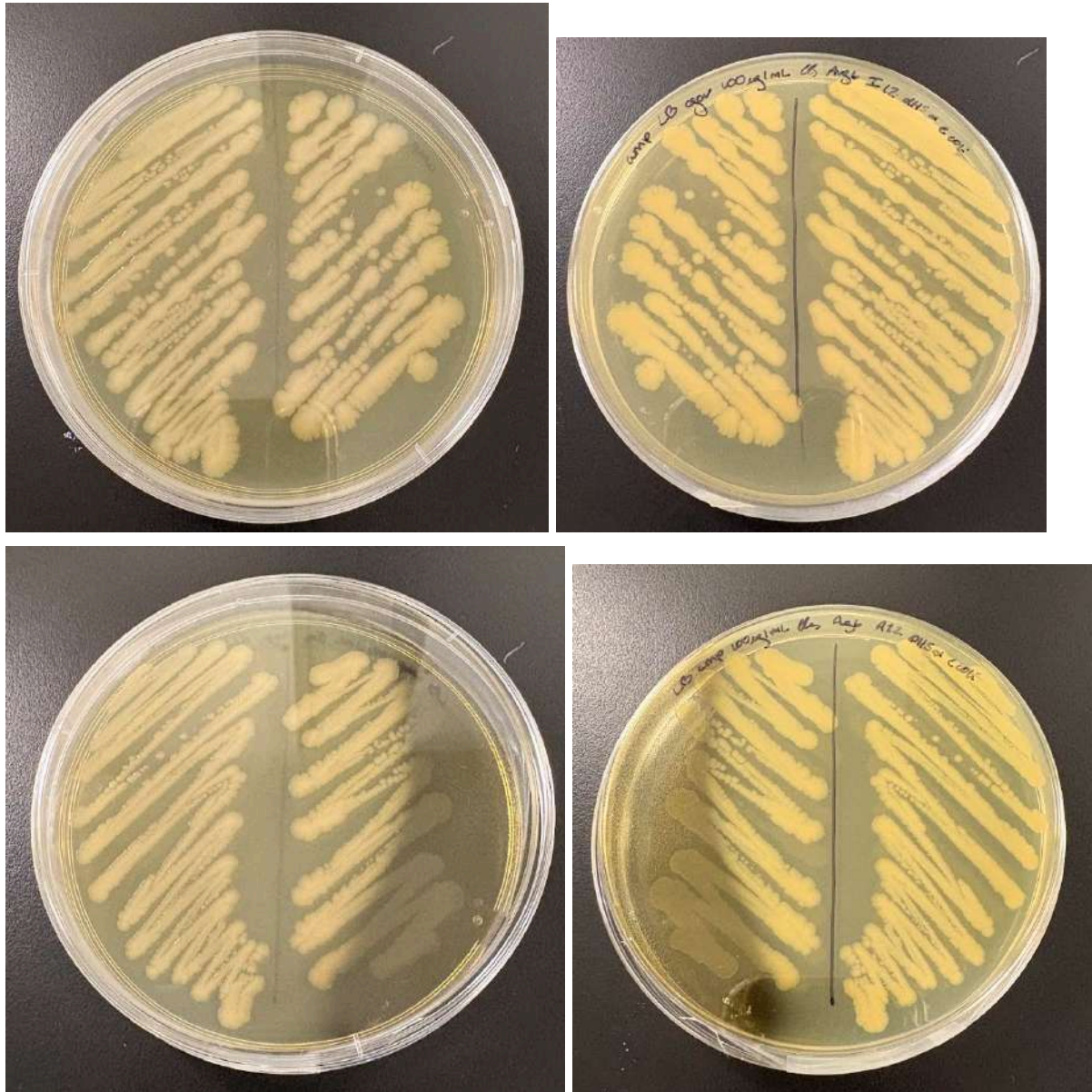


Figure 4. Plates of yeast growth.

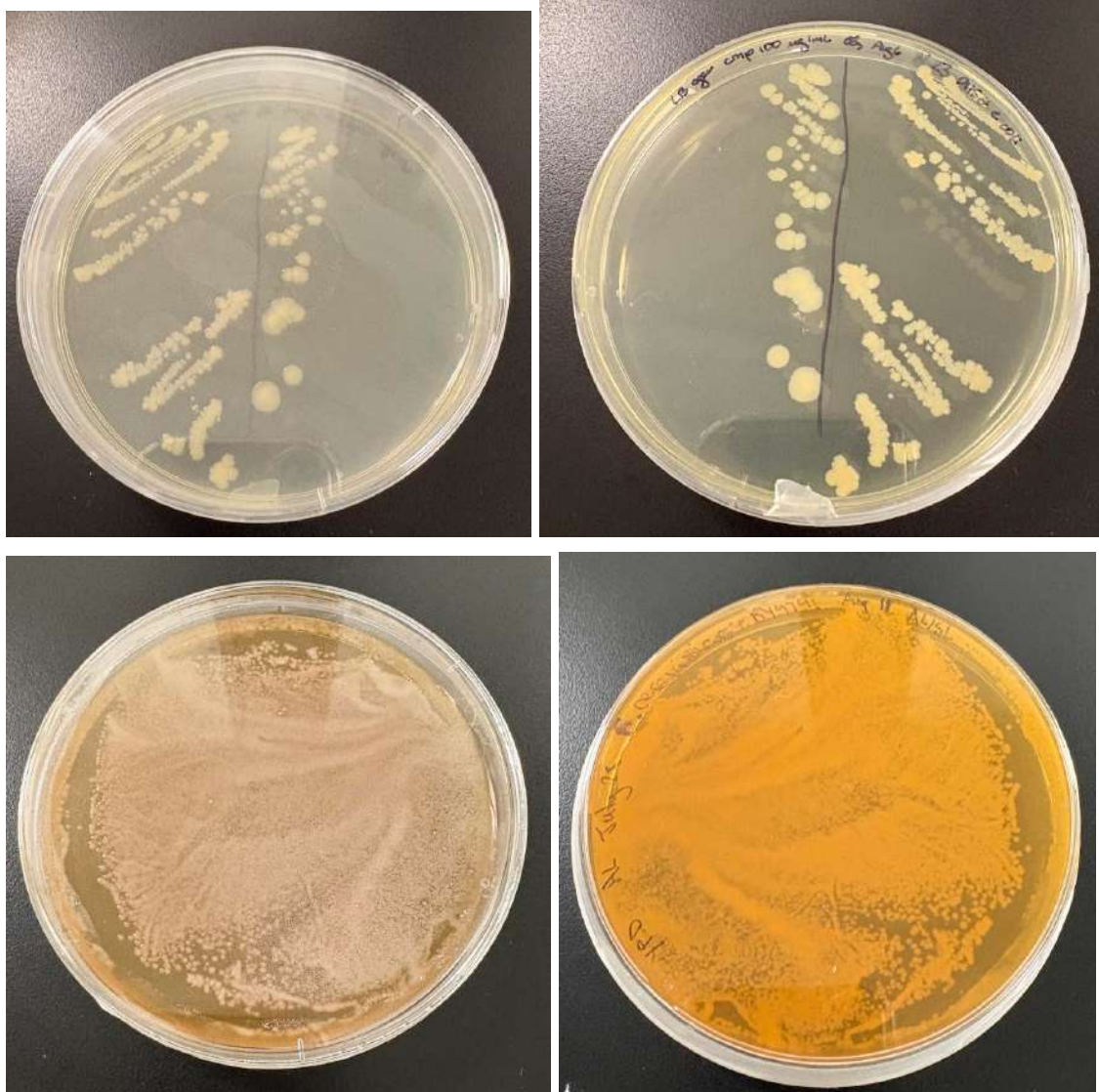


Figure 5. Plates of yeast growth.



Figure 6. Plates of yeast growth.

Next Steps

- Liquid inoculation of transformation + check on yeast transformation

Friday, August 14th

Summary

- Collected REs (dpnl and spel)
 - Sitting in -20
- Gel extraction
 - Made 3 replicates of each, just because each column could only take about a third of total volume
- Mutagenesis with N23 (mscarlet) was done on Sunday
 - Amplification temp was 55 ($T^m = 58$)

Monday, August 17th

Summary

- PCR cleanup for N23
- Making extra LB products for next day

Next Steps

- Transformations

Tuesday, August 18th

Summary

- Camila notes (and todos)
 - Digest N23
 - Quantify N23, RNAs, cpTEF1
 - Make solutions for Rachel to autoclave in morning (LB agar, LB, and centri tubes)
- Transform everything into bacteria
- Make lots of plates (agar media will be prepped) – just for transformations you'll be doing (and one that we'll be doing on Thursday)
- No transformation, nothing is in plasmids, we just use concentrations we have
- Quantify concentration of everything
- Check URA3 plates again

Rna1,2,3, n23,tdh3 ← freezer

Notes

- Did not do transformation because we realized the genes don't have an AMP resistant gene
- Made positive and negative controls for the yeast
 - Took inoculation from fridge, washed it 3 times with nf water, plated on YPD and SD plates, growing in incubator at 30 degrees
- Checked concentration of RNAs and promoter, used elution buffer from gel extraction as the blank, values look interesting...

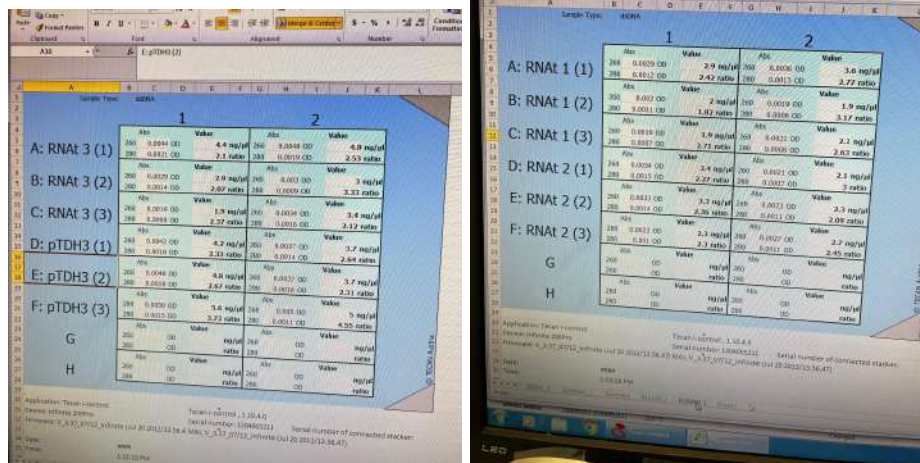


Figure 1. Calculations.

- Did not do n23 because of the confusion of which blank solution to do and we didn't want to waste the samples since we have so little
- Rachel autoclaved 2 bottles of lb agar, 1 empty container for syringe tips, and 1 container of centrifuge tubes
 - We put agar fridge and didn't make more plates because we are running out of fridge space and we have lots of plates already (low on SD plates, only 2 left)

Next Steps

- Check on the control plates in the incubator!

Footnotes

- We are low on competent cells just PSA

Wednesday, August 19th

Summary

- Camila notes
 - Make yeast plates + check that we have enough reagents made for this/next week
 - YPD – at the very least 150 mL
 - Lithium acetate 1M – at least like 5mL
 - Lithium acetate 0.1M – at least like
 - PEG 3350 50% w/v
 - SD plates – 8 minimum, if we wanted to run replicates 12 ;–; but thats probs too much
 - YPD plates – 4 minimum
- Made 9 plates in total with the SD YPD broth
- Melted on the hot plate in a beaker with water
- One of them was moved before it was set, it's a bit lumpy, we'll use that one as a control so it's not difficult to see growth
- Stored in the fridge
- Did a liquid inoculation of the yeast – stored in the incubator for later use
- Checked yesterday's experiment:



Figure 1. YPD plates

- There's growth on the negative control again :(



Figure 2. YPD plates with growth on negative control.

- (Whispy white is growth)

Thursday, August 20th

Summary

- Camila notes
 - Transformed K6 (diff promoter) and GFP dropout plasmid from UofT
 - Transformations were done using the normal protocol (200uL of comp cells, 45s heat shock, 3uL of part)
 - Sitting in bottom incubator at the top row – one negative control and + control made
 - Missed the mScarlet :((may just use the plasmid as is and check concentration
 - Checked to see if we had T4 ligase, T4 ligase buffer, and phosphotases
 - This would be for digest / ligase
 - Apparently DGA1 came in?? Idk we havent gotten txt from Marta rip
 - Workflow for DGA1 + pAN316 looks like this...

DGA1 → digest with BsaI → gel → gel extraction

pAN316 → digest with BsaI → gel → gel extraction

= combine the two and ligate them together

- Phosphotases are needed prior to digest so that the sticky ends don't bind on same plasmid/part
 - Have ordered free sample on creative proteins (likely wont come in time)

Next Steps

- Potentially making SD media + agar again – just to confirm that it isn't anything on our end
 - Rachel can autoclave saturday?

Friday, August 21th

Summary

- Camila notes
 - Made 400mL of SD broth
 - Made 400mL of SD agar
 - Both were using protocols sent to us by Dr Bergey
- Checked plate growth, just noted some on K6 (which makes sense, only 50uL of the transformed bacteria mix was put into the amp plate for GFP dropout plasmid, hoping for growth tomorrow)

Saturday August 22

Summary

- Camila notes
 - Rachel came in to autoclave media (SD media and agar)
 - Noted no growth on GFP dropout plasmid plate

Sunday August 23

Summary

- Camila notes
 - Transformed GFP dropout plasmid again
 - Normal protocol again (200uL of competent cells, 45s of heat shock, 3uL of part, and foregoing positive control to put all of the transformed product into two ampicillin plates)
 - Transformed mutated N23
 - Using normal transformation protocol
 - 200uL comp cells
 - 45s heat shock
 - 3uL of DNA part
 - Inoculated K6 transformed part

Monday, August 24

Summary

- Camila notes
 - Inoculated mutated N23
 - Miniprep K6
 - Transformed GFP dropout again
 - 200uL of competent cells
 - 45s heat shock
 - 4uL of plasmid (all that was left)
 - 2hrs growing in incubator (allowing for more growth)
 - Dispersing all into one plate
 - Inoculated bacteria for comp cells tomorrow

Tuesday, August 25th

Summary

- Camila notes for Andrea and Sabrina
 - From lab store (only buy if you need today)
 - 50mL falcon tubes (check CL1 cupboard if we have)
 - P1000 filtered tips (you will probs need today)
 - Competent cells? I can do it later, just dilute the inoculation I made yesterday
 - Located in bottom incubator, the two furthest falcon tubes
 - Do like 5mL LB and put 50uL bacteria (make sure well mixed)
 - From lab store (only buy if you need today)
 - 50mL falcon tubes (check CL1 cupboard if we have)
 - P1000 filtered tips
 - Make as many amp agar plates you can (ratio of like 3:1 amp lb: lb plates)
 - I think we have two agar bottles, one had a suspension cause it wasnt properly dissolved, try to heat up and see if it still looks funny
 - Put glucose into SD media/agar that was autoclaved the other day (should have my writing on it), you will probs need to heat up sd agar to put in glucose
 - Make SD plates pls! As many as you can with the agar
 - Miniprep N23 pls
- Camila came in to do....
 - Quantify the rest of the DNA
 - Was just missing recently miniprepped N23

	1	2
mN23	Abx 2000 0.0197 OD 200 0.0093 OD	Abx 2000 0.0198 OD 200 0.0093 OD
mN23	Abx 2000 0.0198 OD 200 0.0093 OD	Abx 2000 0.0199 OD 200 0.0093 OD
C	Abx 2000 OD 200 OD	Abx 2000 OD 200 OD
D	Abx 2000 OD 200 OD	Abx 2000 OD 200 OD
E	Abx 2000 OD 200 OD	Abx 2000 OD 200 OD
F	Abx 2000 OD 200 OD	Abx 2000 OD 200 OD
G	Abx 2000 OD 200 OD	Abx 2000 OD 200 OD
H	Abx 2000 OD 200 OD	Abx 2000 OD 200 OD

Figure 1. Computation

- Made abt 15 samples of competent cells, still need to trial puc19 with them
- Golden gate math
- Inoculated yeast for Nile red tomorrow

I know this is a lot, try to prioritize plates and inoculation for comp cells. Miniprep would be nice but not end of world if you don't get to. Thank you!! <33

Notes

- Bought 2 packs of plates from lab store, we already had an unopened p1000 filter tip box, and lots of 50ml tubes so we didn't buy those
- Diluted the inoculation from yesterday for competent cells, currently shaking in incubator
- Made 10 more AMP plates, didn't make lb plates because we have 8 already in the fridge
- Made 14 SD+dextrose plates
 - Made more 40% dextrose (20g dextrose + 50ml water), filter sterilized it, added 20ml to the sd agar
- Miniprep N23 mutagenized and it is in -20 freezer

Footnotes

- Marta says: if we r using UV for BSC, you have to also be there to turn it off because it doesnt automatically turn off
- We are running out of fridge space, was thinking of throwing out some old plates that we dont need...

Wednesday, August 26th

Summary

- Camila notes
 - Today is more chill for you guys (I'll be assembling and transforming in background)
 - I need you to troubleshoot the Nile red experiment. Familiarize with procedure and WHY you are doing each step (we might have to adjust parameters if we don't get good results with the trial today, we need good RELIABLE data)
 - PBS, DMSO, and acetone were brought into CL2 and are sitting on bench
 - Doubtful with the DMSO, pls don't dissolve all of Nile red in it (or like use only what you'll need + a little more today, idk how much Nile red we got)
 - If you get weird results, try again and optimizeeee (my hope is that we get data today from this, and a procedure that is solid for experiments on Sunday/Monday so we aren't panicking)
 - Inoculation of bacteria is sitting in top incubator (I think OD is a part of procedure? Or having nice growing yeast at least)
 - Make 3 gels
 - Make inoculations of BY472 and PXA1
 - Question on this, we definitely don't have enough plasmid to do PXA1 and POX1, so we have to choose one of the two. I wanted to go with PXA1 bc there was more lipid overexpression with that one - (inhibiting transport of acetyl smth smth to peroxisome is better than inhibiting oxidation of acetyl smth once it's in peroxisome no? Even for DGA1 overexpression? I think so)
- What I did
 - Transformed using the competent cells made on Tuesday August 25th
 - Transformed 2x pANS plasmid given to us by Dr Nguyen
 - Transformed 1x pUC19 to validate competent cells
 - One positive and negative control to validate plates/comp cells
 - Golden gate assembly using calculations made on [this](#) document (diluted all of our parts and then combined the following)
 - R1
 - R2
 - R3

- RC
- BG
- BC
- DG
- Ran on thermocycler

37°C	1:00 minute	
16°C	1:00 minute	Repeat step 1-2 29x
60°C	5:00 minutes	
4°C	indefinite	

Stored in -20 after

Notes

- Inoculation of BY4741 and PXA1
 - BY4741
 - Used 500uL of liquid inoculation from last week (RT + 30C) in 2.5mL of YPD
 - PXA1
 - Used plates from passage + 3mL YPD
 - 4 inoculations total (2 each) placed in 30C incubator
- Nile red
 - Prepared 500uL of 2mg/mL Nile red in ACETONE, placed in fridge in Nile Red powder capsule to prevent light from degrading it
 - Used scale in CL2, both only go up to 0.001g and are rated for 0.02 and 0.01g, so accuracy is likely mixed
 - Prepared 1.5mL of DMSO:PBS 1:1 solution (750uL each), tube was warm when mixed??
 - Prepared 1mL of 5ng/mL Nile red in acetone using 30uL 2mg/mL nile red and 970uL acetone
 - Need to make fresh every time but excess is in CL2
 - Used 2mL of overnight culture but OD was only 0.383
 - Used the remaining 2mL of overnight culture and resuspended in 500uL but still only got 0.653 OD
 - Followed protocol and used plate reader, results in results section

- Plate is back in CL2, only used 6 wells
- Protocol changes that probably need to be made:
 - Use 2400RPM for 2min for washing/harvesting
 - ALWAYS harvest ALL OF THE OVERNIGHT CULTURE
 - We can dilute with more PBS if its past 1.0 but it's harder to add cells
 - Ideally need at least 3mL since spectrophotometer uses 2mL
 - Let cells sit for 1-2min after resuspending (apparently reduces background signal according to protocol, need to check protocol cause link isnt letting me access it rn)
- Gels
 - Made 3 agarose gels, stored in SYBR green container (contents had been dumped and container rinsed + filled with TAE) in CL1 fridge
 - Made 1L 1X TAE since we're running low

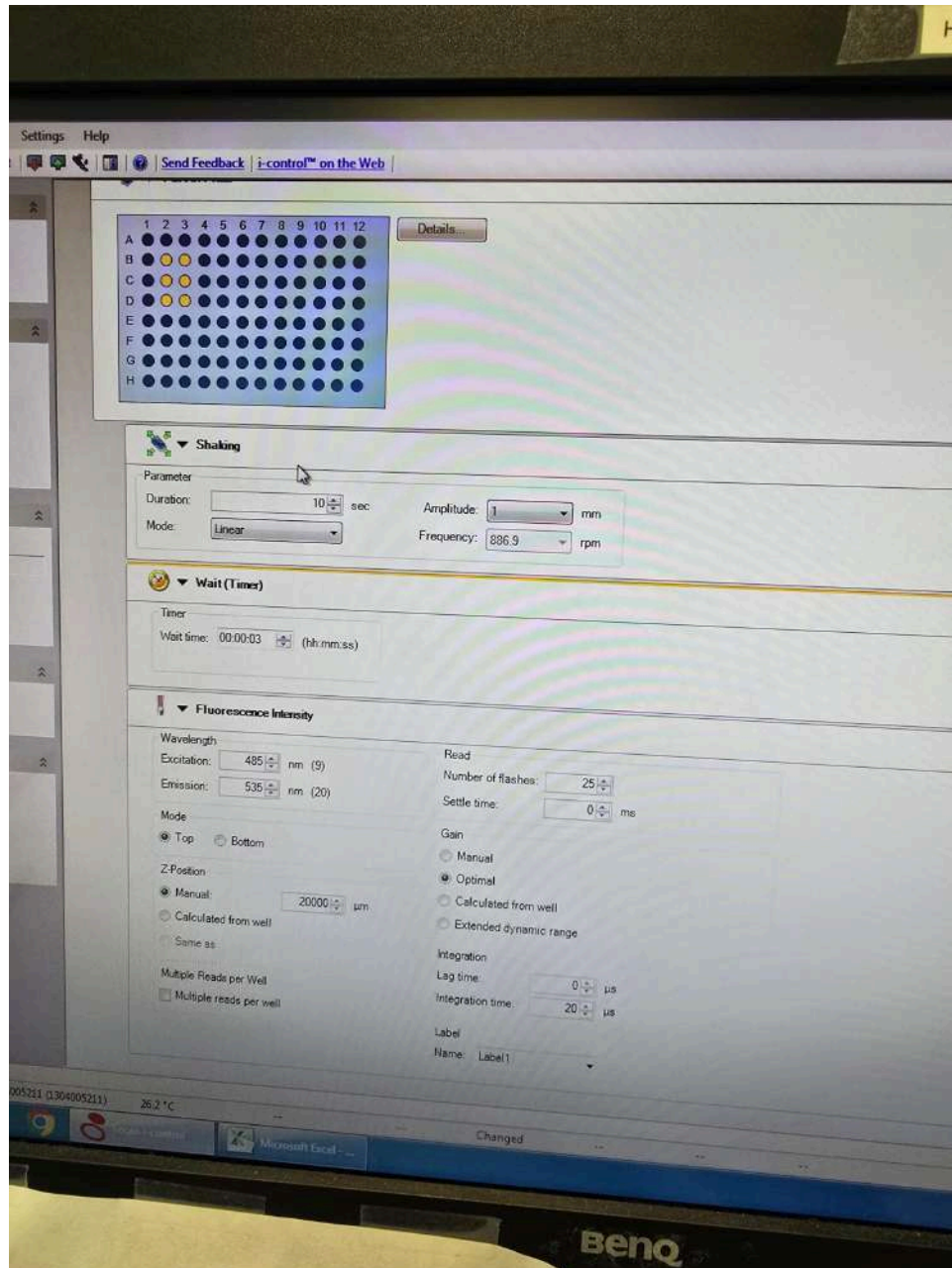


Figure 1. Calculations

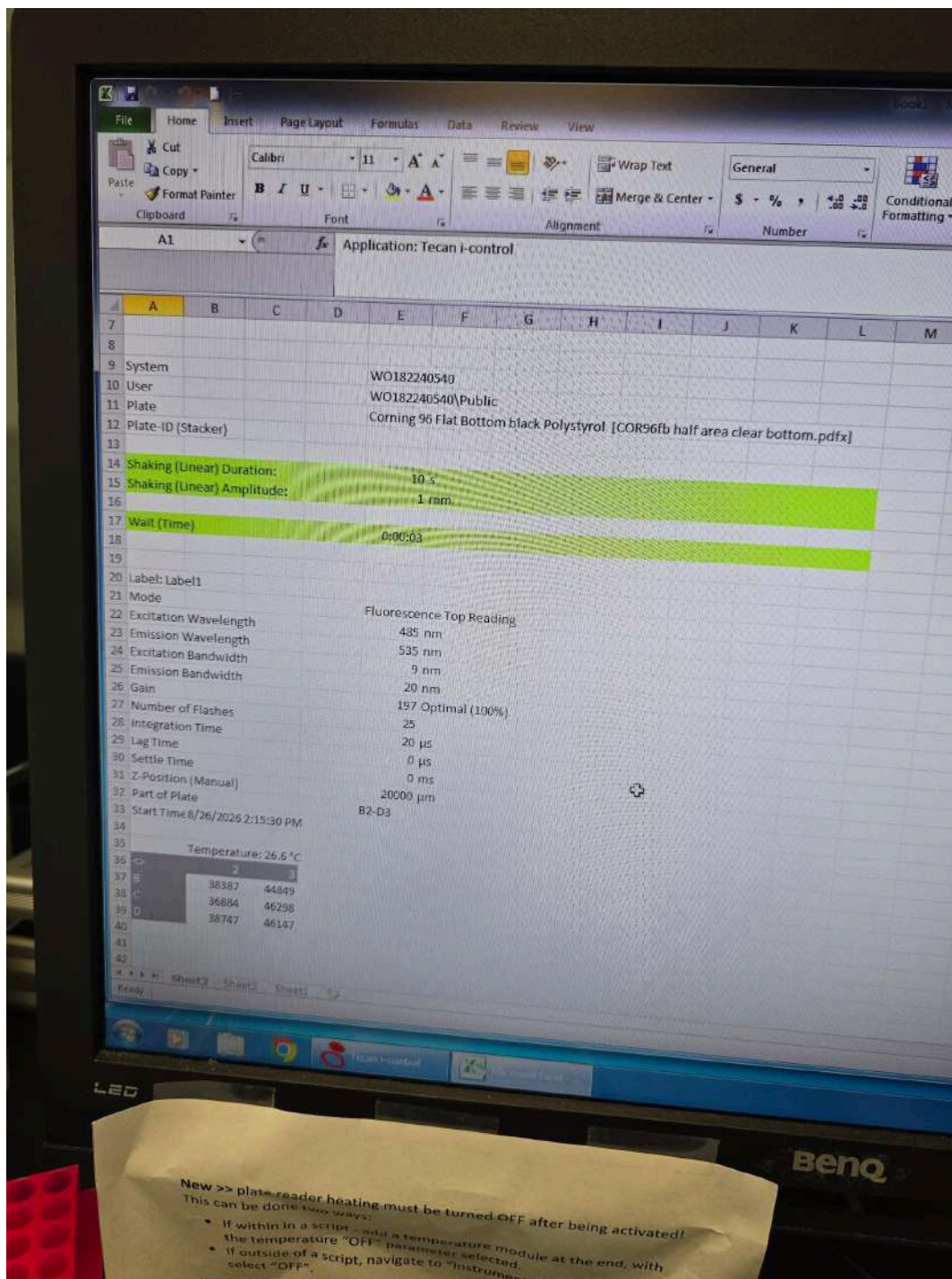


Figure 2. Calculations 2.

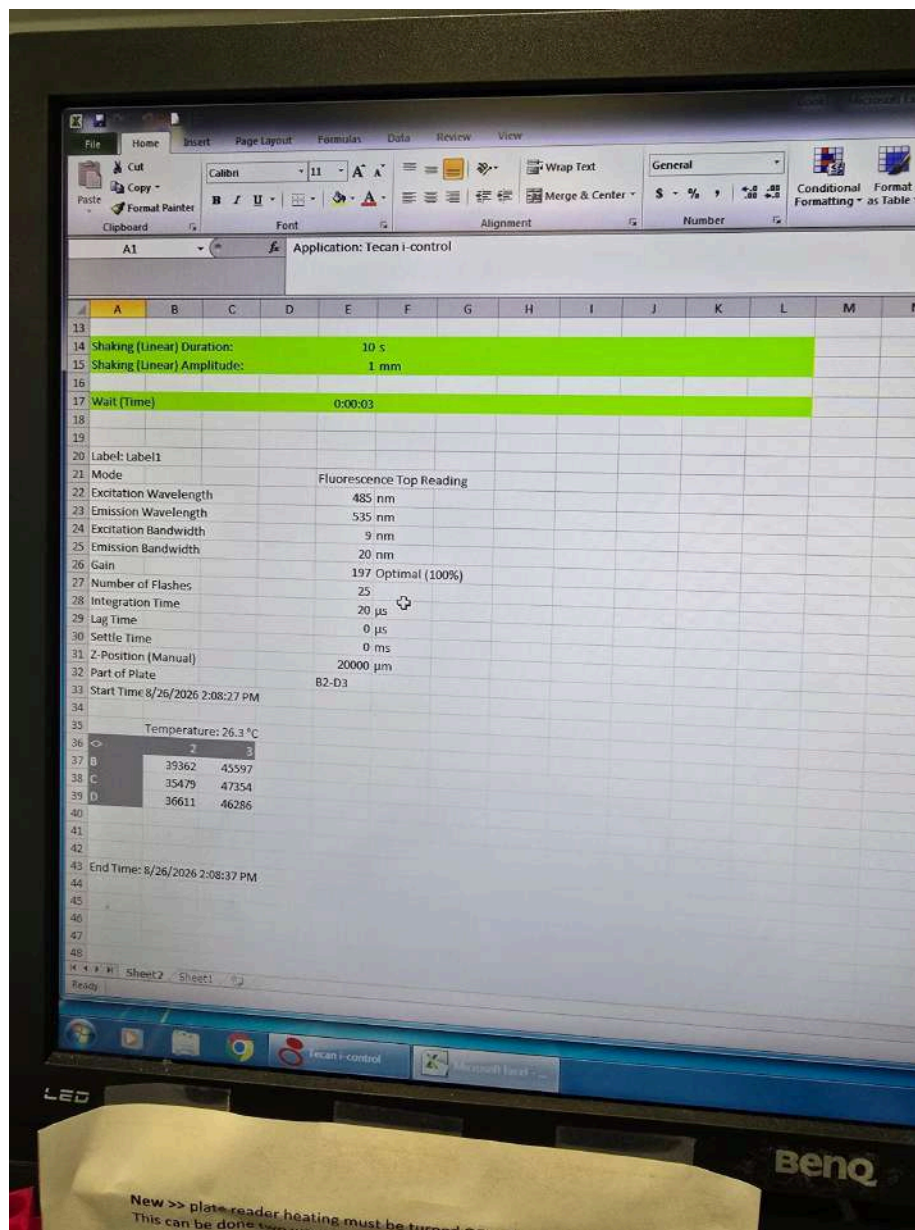


Figure 3. Calculations 3.

Thursday, August 27th

Summary

- Checked plates

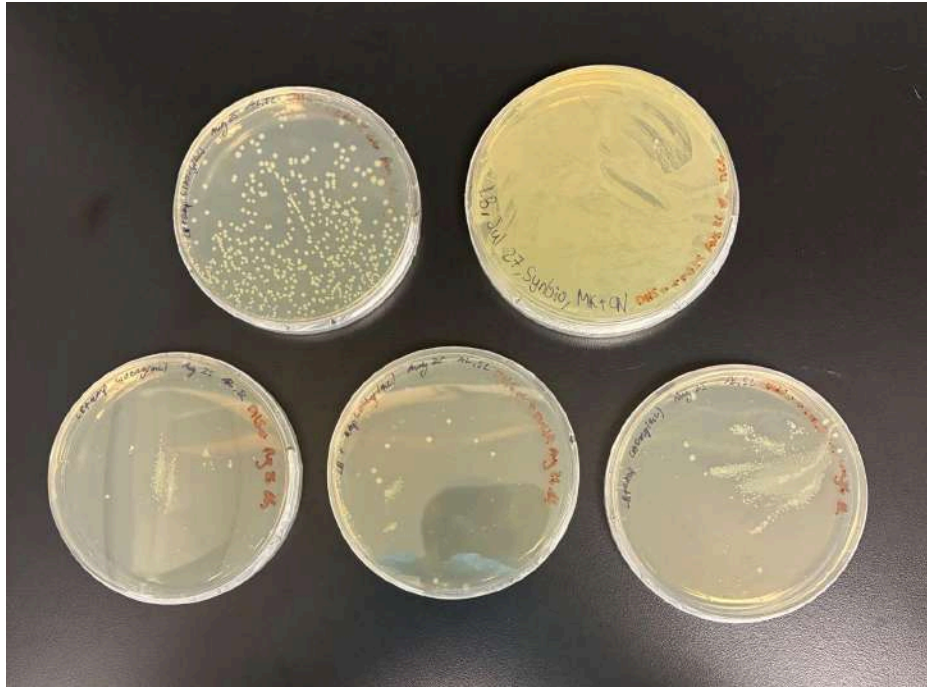


Figure 1. Petri plates.

- Growth on negative plate, no time to run this transformation again
 - Made inoculations of sample from pANS plates (6 from each), 12 colonies selected total
 - Plan will be to digest and run a gel with them and validate which ones were successfully transformed